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Semiconductor device

Abstract

The present disclosure provides a semiconductor device. The semiconductor device includes a substrate, a mounting layer, switching elements, a moisture-resistant layer and a sealing resin. The substrate has a front surface facing in a thickness direction. The mounting layer is electrically conductive and disposed on the front surface. Each switching element includes an element front surface facing in the same direction in which the front surface faces along the thickness direction, a back surface facing in the opposite direction of the element front surface, and a side surface connected to the element front surface and the back surface. The switching elements are electrically bonded to the mounting layer with their back surfaces facing the front surface. The moisture-resistant layer covers at least one side surface. The sealing resin covers the switching elements and the moisture-resistant layer. The moisture-resistant layer is held in contact with the mounting layer and the side surface so as to be spanned between the mounting layer and the side surface in the thickness direction.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a semiconductor device provided with a plurality of switching elements.

BACKGROUND ART

(2) A semiconductor device having a plurality of switching elements such as MOSFET which are electrically-coupled is known. Such a semiconductor device may include a case made of a synthetic

resin and a wiring board supported by the case. The switching elements are electrically connected to the wiring board. The case and the wiring board surrounds a space, which may be filled with a sealing resin such as silicone gel. The switching elements are covered with the sealing resin.

(3) In recent years, semiconductor devices with relatively high rating voltages are increasingly demanded in areas with tropical climate near/on the equator. In tropical areas, semiconductor devices are placed in a high temperature and high humidity environment. For a semiconductor device to operate stably in such an environment, it is desirable that the semiconductor device passes the H3TRB (High Humidity High Temperature Reverse Bias) test. The H3TRB test estimates the withstand time (unit: hours) of a semiconductor device when it is driven at 80% of its rated DC voltage under high temperature and high humidity conditions (temperature: 85° C., humidity: 85%). In the H3TRB test, semiconductor devices with withstand time of 1000 h or more are considered as acceptable. The semiconductor devices accepted by this test are expected to operate stably under high temperature and high humidity conditions.

(4) By performing the H3TRB test on the above-described semiconductor devices, the inventors have found that the withstand time of the devices is highly likely to be short of 1000 h. If moisture enters the sealing resin of a semiconductor device placed under high temperature and high humidity conditions, the dielectric breakdown voltage of the sealing resin will deteriorate, which may allow leak current in the switching elements. If the leak current reaches the wiring board, at least one of the switching elements may be destroyed, and consequently, the withstand time of the device will be shortened. With a higher rating voltage, a semiconductor device tends to have a shorter withstand time. Thus, in order to operate stably under high temperature and high humidity conditions, a semiconductor device may need to pass, as a criterion, the H3TRB test for the desired rating voltage.

SUMMARY OF THE INVENTION

(5) In light of the above circumstances, the present disclosure aims to provide a semiconductor device capable of operating stably under high temperature and high humidity conditions.

(6) According to the present disclosure, a semiconductor device is provided. The semiconductor device includes a substrate, a mounting layer, switching elements, a moisture-resistant layer and a sealing resin. The substrate has a front surface facing in a thickness direction. The mounting layer is electrically conductive and disposed on the front surface. Each of the switching elements includes an element front surface facing in a same direction in which the front surface faces along the thickness direction, a back surface facing in the opposite direction of the element front surface, and a side surface connected to both of the element front surface and the back surface. The switching elements are electrically bonded to the mounting layer with the back surface facing the front surface. The moisture-resistant layer covers at least one of the side surfaces. The sealing resin covers both of the switching elements and the moisture-resistant layer. The moisture-resistant layer is held in contact with both of the mounting layer and the side surface so as to be spanned between the mounting layer and the side surface in the thickness direction.

(7) Other features and advantages of the present disclosure will become apparent from the following detailed description with reference to the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of a semiconductor device according to a first embodiment of the present disclosure;

(2) FIG. 2 is a plan view of the semiconductor device shown in FIG. 1;

(3) FIG. 3 is a plan view of the semiconductor device shown in FIG. 1 (as seen through a sealing resin, a moisture-resistant layer and a top plate);

(4) FIG. 4 is a front view of the semiconductor device shown in FIG. 1;
(5) FIG. 5 is a right side view of the semiconductor device shown in FIG. 1;
(6) FIG. 6 is a left side view of the semiconductor device shown in FIG. 1;
(7) FIG. 7 is a bottom view of the semiconductor device shown in FIG. 1;
(8) FIG. 8 shows a right portion (near the first substrate) of FIG. 3 as enlarged;
(9) FIG. 9 shows a left portion (near the second substrate) of FIG. 3 as enlarged;
(10) FIG. 10 shows a central portion (near the third substrate) of FIG. 3 as enlarged;
(11) FIG. 11 is a sectional view taken along line XI-XI in FIG. 3;
(12) FIG. 12 is a sectional view taken along line XII-XII in FIG. 3;
(13) FIG. 13 is a sectional view taken along line XIII-XIII in FIG. 3;
(14) FIG. 14 is a sectional view taken along line XIV-XIV in FIG. 3;
(15) FIG. 15 shows a portion of FIG. 3 (a switching element and a protective element bonded to an upper arm mounting layer) as enlarged;
(16) FIG. 16 is a sectional view taken along line XVI-XVI in FIG. 15;
(17) FIG. 17 is a sectional view taken along line XVII-XVII in FIG. 15;
(18) FIG. 18 shows a portion of FIG. 3 (a switching element and a protective element bonded to a lower arm mounting layer) as enlarged;
(19) FIG. 19 is a sectional view taken along line XIX-XIX in FIG. 18;
(20) FIG. 20 is a sectional view taken along line XX-XX in FIG. 18;
(21) FIG. 21 is a circuit diagram of the semiconductor device shown in FIG. 1;
(22) FIG. 22 shows a portion of FIG. 16 as enlarged;
(23) FIG. 23 is a sectional view showing a portion of a semiconductor device of a comparative example (a switching element bonded to an upper arm layer) as enlarged;
(24) FIG. 24 is a sectional view showing a portion of a semiconductor device (a switching element and a protective element bonded to the upper arm layer) according to a first variation of the first embodiment of the present disclosure;
(25) FIG. 25 is a sectional view showing a portion of the semiconductor device (a switching element and a protective element bonded to the lower arm layer) according to the first variation of the first embodiment of the present disclosure;
(26) FIG. 26 is a plan view showing a portion of a semiconductor device (a switching element and a protective element bonded to the upper arm mounting layer) according to a second variation of the first embodiment of the present disclosure;
(27) FIG. 27 is a sectional view taken along line XXVII-XXVII in FIG. 26;
(28) FIG. 28 is a sectional view taken along line XXVIII-XXVIII in FIG. 26;
(29) FIG. 29 is a plan view showing a portion of a semiconductor device (a switching element and a protective element bonded to the lower arm mounting layer) according to the second variation of the first embodiment of the present disclosure;
(30) FIG. 30 is a sectional view taken along line XXX-XXX in FIG. 29;
(31) FIG. 31 is a sectional view taken along line XXXI-XXXI in FIG. 29;
(32) FIG. 32 is a plan view showing a portion of a semiconductor device (a switching element and a protective element bonded to the upper arm mounting layer) according to a third variation of the first embodiment of the present disclosure;
(33) FIG. 33 is a sectional view taken along line XXXIII-XXXIII in FIG. 32;
(34) FIG. 34 is a sectional view taken along line XXXIV-XXXIV in FIG. 32;
(35) FIG. 35 is a plan view showing a portion of a semiconductor device (a switching element and a protective element bonded to the lower arm mounting layer) according to the third variation of the first embodiment of the present disclosure;
(36) FIG. 36 is a sectional view taken along line XXXVI-XXXVI in FIG. 35;
(37) FIG. 37 is a sectional view taken along line XXXVII-XXXVII in FIG. 35;
(38) FIG. 38 is a plan view showing a portion of a semiconductor device (a switching element and

a protective element bonded to the upper arm mounting layer) according to a fourth variation of the first embodiment of the present disclosure;

(39) FIG. **39** is a sectional view taken along line XXXIX-XXXIX in FIG. **38**;

(40) FIG. **40** is a sectional view taken along line XL-XL in FIG. **38**;

(41) FIG. **41** is a plan view showing a portion of a semiconductor device (a switching element and a protective element bonded to the lower arm mounting layer) according to the fourth variation of the first embodiment of the present disclosure;

(42) FIG. **42** is a sectional view taken along line XLII-XLII in FIG. **41**;

(43) FIG. **43** is a sectional view taken along line XLIII-XLIII in FIG. **41**;

(44) FIG. **44** is a plan view showing a portion of a semiconductor device (a switching element and a protective element bonded to the upper arm mounting layer) according to a fifth variation of the first embodiment of the present disclosure;

(45) FIG. **45** is a sectional view taken along line XLV-XLV in FIG. **44**;

(46) FIG. **46** is a sectional view taken along line XLVI-XLVI in FIG. **45**;

(47) FIG. **47** is a plan view showing a portion of the semiconductor device (a switching element and a protective element bonded to the lower arm mounting layer) according to the fifth variation of the first embodiment of the present disclosure;

(48) FIG. **48** is a sectional view taken along line XLVIII-XLVIII in FIG. **47**;

(49) FIG. **49** is a sectional view taken along line XLIX-XLIX in FIG. **47**;

(50) FIG. **50** shows test results based on variations in thicknesses of the moisture-resistant layer of the semiconductor device according to the fourth variation of the first embodiment of the present disclosure;

(51) FIG. **51** shows results of the H3TRB test on the semiconductor device according to the fourth variation of the first embodiment of the present disclosure and the semiconductor device of the comparative example;

(52) FIG. **52** is a plan view showing a portion of a semiconductor device (a switching element and a protective element bonded to the upper arm mounting layer) according to a second embodiment of the present disclosure;

(53) FIG. **53** is a sectional view taken along line LIII-LIII in FIG. **52**;

(54) FIG. **54** is a sectional view taken along line LIV-LIV in FIG. **52**;

(55) FIG. **55** is a plan view showing a portion of the semiconductor device (a switching element and a protective element bonded to the lower arm mounting layer) according to the second embodiment of the present disclosure;

(56) FIG. **56** is a sectional view taken along line LVI-LVI in FIG. **55**; and

(57) FIG. **57** is a sectional view taken along line LVII-LVII in FIG. **55**.

MODE FOR CARRYING OUT THE INVENTION

(58) Modes for carrying out the disclosure (hereinafter referred to as embodiments) are described below with reference to the accompanying drawings.

First Embodiment

(59) With reference to FIGS. **1** to **23**, a semiconductor device **A10** according to a first embodiment of the present disclosure is described. The semiconductor device **A10** includes a substrate **11**, a first mounting layer **211**, a second mounting layer **221**, a third mounting layer **231**, switching elements **31**, a moisture-resistant layer **51** and a sealing resin **52**. Of these, the first mounting layer **211**, the second mounting layer **221** and the third mounting layer **231** are examples of the “mounting layer” as set forth in the appended claims of the present disclosure. In addition to these, the semiconductor device **A10** further includes a first electroconductive layer **212**, a second electroconductive layer **222**, a third electroconductive layer **232**, a power supply terminal **24**, an output terminal **25**, a connecting electroconductive member **261**, protective elements **32**, wires **41**, a heat sink **61** and a case **70**. Of these, the first electroconductive layer **212**, the second electroconductive layer **222** and the third electroconductive layer **232** are examples of the “electroconductive layer” as set forth in

the appended claims of the present disclosure. The power supply terminal **24** includes a first power supply terminal **24A** and a second power supply terminal **24B**. For easier understanding, FIG. **3** shows a view seen through the moisture-resistant layer **51**, the sealing resin **52** and a top plate **79**. In FIG. **3**, the line XI-XI and the line XII-XII are indicated by dash-dotted lines. In FIGS. **11** and **12**, illustration of the moisture-resistant layer **51** is omitted.

(60) The semiconductor device **A10** shown in FIG. **1** is a power module. The semiconductor device **A10** may be used for inverter devices of various electric products. As shown in FIGS. **1** and **2**, the semiconductor device **A10** is rectangular as viewed in the thickness direction z of the substrate **11**. For convenience of explanation, a direction that is perpendicular to the thickness direction z of the substrate **11** (hereinafter simply “thickness direction z ”) is referred to as the “first direction $x1$ ”. The direction that is perpendicular to both of the thickness direction z and the first direction $x1$ is referred to as the “second direction $x2$ ”. The longitudinal direction of the semiconductor device **A10** is the second direction $x2$.

(61) The substrate **11** is an electrically insulating member on which the mounting layer (the first mounting layer **211**, the second mounting layer **221** and the third mounting layer **231**) and the electroconductive layer (the first electroconductive layer **212**, the second electroconductive layer **222** and the third electroconductive layer **232**) are disposed, as shown in FIG. **3**. The substrate **11** has three sections, namely a first substrate **11A**, a second substrate **11B** and a third substrate **11C**. The first substrate **11A**, the second substrate **11B** and the third substrate **11C** are spaced apart from each other in the second direction $x2$. In the second direction $x2$, the third substrate **11C** is located between the first substrate **11A** and the second substrate **11B**. Unlike this configuration, the substrate **11** may have two sections, namely the first substrate **11A** and the second substrate **11B**, or may only have a single section. As shown in FIG. **11**, each of the first substrate **11A**, the second substrate **11B** and the third substrate **11C** has a front surface **111** and a back surface **112** facing away from each other in the thickness direction z .

(62) The substrate **11** is made of a ceramic with excellent thermal conductivity. Examples of such a ceramic include aluminum nitride (AlN). A DBC (Direct Bonding Copper) substrate, which has copper (Cu) foils bonded to the front surface **111** and the back surface **112**, may be used as the substrate **11**. By using a DBC substrate, the mounting layer and the electroconductive layer can be easily formed through patterning of the copper foil bonded to the front surface **111**. The copper foil bonded to the back surface **112** can be formed into a heat transfer layer **62** (described later).

(63) As shown in FIGS. **3** and **8**, on the front surface **111** of the first substrate **11A** are disposed the first mounting layer **211**, the first electroconductive layer **212**, a first gate layer **213**, a first detection layer **214** and a thermistor mounting layer **215**. These are electroconductive members made of a thin metal film such as a copper foil. The surfaces of these layers may be plated with silver (Ag), for example.

(64) As shown in FIG. **8**, switching elements **31** and protective elements **32** are electrically bonded to the first mounting layer **211**. The first mounting layer **211** includes a first upper arm mounting layer **211A** and a first lower arm mounting layer **211B**.

(65) As shown in FIG. **8**, the first upper arm mounting layer **211A** is offset toward one end of the first substrate **11A** (upper side in FIG. **8**) in the first direction $x1$. The first upper arm mounting layer **211A** is in the form of a strip extending along the second direction $x2$. Three switching elements **31** and three protective elements **32** are electrically bonded to the first upper arm mounting layer **211A**. Note that the number of switching elements **31** and the number of the protective elements **32** to be electrically bonded to the first upper arm mounting layer **211A** are not limited to three. On the first upper arm mounting layer **211A**, both of the switching elements **31** and the protective elements **32** are aligned in the second direction $x2$. The first upper arm mounting layer **211A** is formed with a first power supply pad **211C** in the form of a strip extending along the first direction $x1$ at an end close to the case **70** in the second direction $x2$. The first power supply pad **211C** is electrically connected to the first power supply terminal **24A**.

(66) As shown in FIG. 8, the first lower arm mounting layer **211B** is located between the first upper arm mounting layer **211A** and the first electroconductive layer **212** in the first direction **x1**. The first lower arm mounting layer **211B** is in the form of a strip extending along the second direction **x2**. Three switching elements **31** and three protective elements **32** are electrically bonded to the first lower arm mounting layer **211B**. Note that the number of switching elements **31** and the number of the protective elements **32** to be electrically bonded to the first lower arm mounting layer **211B** are not limited to three. On the first lower arm mounting layer **211B**, both of the switching elements **31** and the protective elements **32** are aligned in the second direction **x2**. As shown in FIG. 15, the first lower arm mounting layer **211B** is electrically connected, via wires **41**, to both of the front surface electrodes **311** (described later) of the switching elements **31** and anode electrodes **321** (described later) of the protective elements **32** that are electrically bonded to the first upper arm mounting layer **211A**.

(67) As shown in FIGS. 8 and 18, the first electroconductive layer **212** is electrically connected, via wires **41**, to the front surface electrodes **311** of the switching elements **31** and the anode electrodes **321** of the protective elements **32** that are electrically bonded to the first lower arm mounting layer **211B**. The first electroconductive layer **212** is offset toward the other end of the first substrate **11A** (lower side in FIG. 8) in the first direction **x1**. The first electroconductive layer **212** is in the form of a strip extending along the second direction **x2**. The first electroconductive layer **212** is formed with a second power supply pad **212A** in the form of a strip extending along the first direction **x1** at an end close to the case **70** in the second direction **x2**. The second power supply pad **212A** is electrically connected to the second power supply terminal **24B**.

(68) As shown in FIGS. 15 and 18, the first gate layer **213** is electrically connected, via first gate wires **421**, to gate electrodes **313** (described later) of the switching elements **31** electrically bonded to the first mounting layer **211**. The first gate layer **213** is in the form of a strip extending along the second direction **x2** and faces the switching elements **31** as viewed in the thickness direction **z**. The first gate layer **213** includes a first upper arm gate layer **213A** and a first lower arm gate layer **213B**.

(69) As shown in FIG. 8, the first upper arm gate layer **213A** is located between the first upper arm mounting layer **211A** and the case **70** in the first direction **x1**. As viewed in the thickness direction **z**, the first upper arm gate layer **213A** faces the switching elements **31** electrically bonded to the first upper arm mounting layer **211A**. As shown in FIG. 15, the first upper arm gate layer **213A** is electrically connected, via first gate wires **421**, to the gate electrodes **313** of the switching elements **31** electrically bonded to the first upper arm mounting layer **211A**.

(70) As shown in FIG. 8, the first lower arm gate layer **213B** is located between the first lower arm mounting layer **211B** and the first electroconductive layer **212** in the first direction **x1**. As viewed in the thickness direction **z**, the first lower arm gate layer **213B** faces the switching elements **31** electrically bonded to the first lower arm mounting layer **211B**. As shown in FIG. 18, the first lower arm gate layer **213B** is electrically connected, via first gate wires **421**, to the gate electrodes **313** of the switching elements **31** electrically bonded to the first lower arm mounting layer **211B**.

(71) As shown in FIGS. 15 and 18, the first detection layer **214** is electrically connected, via first detection wires **431**, to the front surface electrodes **311** of the switching elements **31** electrically bonded to the first mounting layer **211**. The first detection layer **214** is in the form of a strip extending along the second direction **x2** and faces the switching elements **31** as viewed in the thickness direction **z**. The first detection layer **214** includes a first upper arm detection layer **214A** and a first lower arm detection layer **214B**.

(72) As shown in FIG. 8, the first upper arm detection layer **214A** is located between the first upper arm mounting layer **211A** and the first upper arm gate layer **213A** in the first direction **x1**. As viewed in the thickness direction **z**, the first upper arm detection layer **214A** faces the switching elements **31** electrically bonded to the first upper arm mounting layer **211A**. As shown in FIG. 15, the first upper arm detection layer **214A** is electrically connected, via first detection wires **431**, to

the front surface electrodes **311** of the switching elements **31** electrically bonded to the first upper arm mounting layer **211A**.

(73) As shown in FIG. **8**, the first lower arm detection layer **214B** is located between the first lower arm mounting layer **211B** and the first lower arm gate layer **213B** in the first direction **x1**. The first lower arm detection layer **214B** is in the form of an L-shaped strip with a part extending in the first direction **x1** and a part extending in the second direction **x2**. As viewed in the thickness direction **z**, the part extending in the second direction **x2** faces the switching elements **31** electrically bonded to the first lower arm mounting layer **211B**. As shown in FIG. **18**, the first lower arm detection layer **214B** is electrically connected, via first detection wires **431**, to the front surface electrodes **311** of the switching elements **31** electrically bonded to the first lower arm mounting layer **211B**.

(74) As shown in FIG. **8**, a thermistor **33** is electrically bonded to the thermistor mounting layer **215**. The thermistor mounting layer **215** is located close to a corner of the first substrate **11A**. The thermistor mounting layer **215** is surrounded by the first upper arm mounting layer **211A**, the first upper arm gate layer **213A** and the first upper arm detection layer **214A**. The thermistor mounting layer **215** has a pair of sections spaced apart from each other in the second direction **x2**. The positive electrode of the thermistor **33** is electrically bonded to one of these sections, whereas the negative electrode of the thermistor **33** is electrically bonded to the other one of these sections.

(75) As shown in FIGS. **3** and **9**, on the front surface **111** of the second substrate **11B** are disposed the second mounting layer **211**, the second electroconductive layer **222**, a second gate layer **223** and a second detection layer **224**. These are electroconductive members made of a thin metal film such as a copper foil. The surfaces of these layers may be plated with silver, for example.

(76) As shown in FIG. **9**, switching elements **31** and protective elements **32** are electrically bonded to the second mounting layer **221**. The second mounting layer **221** includes a second upper arm mounting layer **221A** and a second lower arm mounting layer **221B**.

(77) As shown in FIG. **9**, the second upper arm mounting layer **221A** is offset toward one end of the second substrate **11B** (upper side in FIG. **9**) in the first direction **x1**. The second upper arm mounting layer **221A** is in the form of a strip extending along the second direction **x2**. Three switching elements **31** and three protective elements **32** are electrically bonded to the second upper arm mounting layer **221A**. Note that the number of switching elements **31** and the number of the protective elements **32** to be electrically bonded to the second upper arm mounting layer **221A** are not limited to three. On the second upper arm mounting layer **221A**, both of the switching elements **31** and the protective element **32** are aligned in the second direction **x2**.

(78) As shown in FIG. **9**, the second lower arm mounting layer **221B** is located between the second upper arm mounting layer **221A** and the second electroconductive layer **222** in the first direction **x1**. The second lower arm mounting layer **221B** is in the form of a strip extending along the second direction **x2**. Three switching elements **31** and three protective elements **32** are electrically bonded to the second lower arm mounting layer **221B**. Note that the number of switching elements **31** and the number of the protective elements **32** to be electrically bonded to the second lower arm mounting layer **221B** are not limited to three. On the second lower arm mounting layer **221B**, both of the switching elements **31** and the protective element **32** are aligned in the second direction **x2**. As shown in FIG. **15**, the second lower arm mounting layer **221B** is electrically connected, via wires **41**, to the front surface electrodes **311** of the switching elements **31** and the anode electrodes **321** of the protective elements **32** that are electrically bonded to the second upper arm mounting layer **221A**. The second lower arm mounting layer **221B** is formed with an output pad **221C** in the form of a strip extending along the first direction **x1** at an end close to the case **70** in the second direction **x2**. In the second direction **x2**, the output pad **221C** is close to both of the second upper arm mounting layer **221A** and the second electroconductive layer **222**. The output pad **221C** is electrically connected to the output terminal **25**.

(79) As shown in FIGS. **9** and **18**, the second electroconductive layer **222** is electrically connected, via wires **41**, to the front surface electrodes **311** of the switching elements **31** and the anode

electrodes **321** of the protective elements **32** that are electrically bonded to the second lower arm mounting layer **221B**. The second electroconductive layer **222** is offset toward the other end of the second substrate **11B** (lower side in FIG. **9**) in the first direction **x1**. The second electroconductive layer **222** is in the form of a strip extending along the second direction **x2**.

(80) As shown in FIGS. **15** and **18**, the second gate layer **223** is electrically connected, via first gate wires **421**, to the gate electrodes **313** of the switching elements **31** electrically bonded to the second mounting layer **221**. The second gate layer **223** is in the form of a strip extending along the second direction **x2** and faces the switching elements **31** as viewed in the thickness direction **z**. The second gate layer **223** includes a second upper arm gate layer **223A** and a second lower arm gate layer **223B**.

(81) As shown in FIG. **9**, the second upper arm gate layer **223A** is located between the second upper arm mounting layer **221A** and the case **70** in the first direction **x1**. As viewed in the thickness direction **z**, the second upper arm gate layer **223A** faces the switching elements **31** electrically bonded to the second upper arm mounting layer **221A**. As shown in FIG. **15**, the second upper arm gate layer **223A** is electrically connected, via first gate wires **421**, to the gate electrodes **313** of the switching elements **31** electrically bonded to the second upper arm mounting layer **221A**.

(82) As shown in FIG. **9**, the second lower arm gate layer **223B** is located between the second lower arm mounting layer **221B** and the second electroconductive layer **222** in the first direction **x1**. As viewed in the thickness direction **z**, the second lower arm gate layer **223B** faces the switching elements **31** electrically bonded to the second lower arm mounting layer **221B**. As shown in FIG. **18**, the second lower arm gate layer **223B** is electrically connected, via first gate wires **421**, to the gate electrodes **313** of the switching elements **31** electrically bonded to the second lower arm mounting layer **221B**.

(83) As shown in FIGS. **15** and **18**, the second detection layer **224** is electrically connected, via first detection wires **431**, to the front surface electrodes **311** of the switching elements **31** electrically bonded to the second mounting layer **221**. The second detection layer **224** is in the form of a strip extending along the second direction **x2** and faces the switching elements **31** as viewed in the thickness direction **z**. The second detection layer **224** includes a second upper arm detection layer **224A** and a second lower arm detection layer **224B**.

(84) As shown in FIG. **9**, the second upper arm detection layer **224A** is located between the second upper arm mounting layer **221A** and the second upper arm gate layer **223A** in the first direction **x1**. As viewed in the thickness direction **z**, the second upper arm detection layer **224A** faces the switching elements **31** electrically bonded to the second upper arm mounting layer **221A**. As shown in FIG. **15**, the second upper arm detection layer **224A** is electrically connected, via first detection wires **431**, to the front surface electrodes **311** of the switching elements **31** electrically bonded to the second upper arm mounting layer **221A**.

(85) As shown in FIG. **9**, the second lower arm detection layer **224B** is located between the second lower arm mounting layer **221B** and the second lower arm gate layer **223B** in the first direction **x1**. As viewed in the thickness direction **z**, the second lower arm detection layer **224B** faces the switching elements **31** electrically bonded to the second lower arm mounting layer **221B**. As shown in FIG. **18**, the second lower arm detection layer **224B** is electrically connected, via first detection wires **431**, to the front surface electrodes **311** of the switching elements **31** electrically bonded to the second lower arm mounting layer **221B**.

(86) As shown in FIGS. **3** and **10**, on the front surface **111** of the third substrate **11C** are disposed the third mounting layer **231**, the third electroconductive layer **232**, a third gate layer **233** and a third detection layer **234**. These are electroconductive members made of a thin metal film such as a copper foil. The surfaces of these layers may be plated with silver, for example.

(87) As shown in FIG. **10**, switching elements **31** and protective elements **32** are electrically bonded to the third mounting layer **231**. The third mounting layer **231** includes a third upper arm mounting layer **231A** and a third lower arm mounting layer **231B**.

(88) As shown in FIG. 10, the third upper arm mounting layer **231A** is offset toward one end of the third substrate **11C** (upper side in FIG. 10) in the first direction **x1**. The third upper arm mounting layer **231A** is in the form of a strip extending along the second direction **x2**. Two switching elements **31** and two protective elements **32** are electrically bonded to the third upper arm mounting layer **231A**. Note that the number of switching elements **31** and the number of the protective elements **32** to be electrically bonded to the third upper arm mounting layer **231A** are not limited to two. On the third upper arm mounting layer **231A**, both of the switching elements **31** and the protective element **32** are aligned in the second direction **x2**.

(89) As shown in FIG. 10, the third lower arm mounting layer **231B** is located between the third upper arm mounting layer **231A** and the third electroconductive layer **232** in the first direction **x1**. The third lower arm mounting layer **231B** is in the form of a strip extending along the second direction **x2**. Two switching elements **31** and two protective elements **32** are electrically bonded to the third lower arm mounting layer **231B**. Note that the number of switching elements **31** and the number of the protective element **32** to be electrically bonded to the third lower arm mounting layer **231B** are not limited to two. On the third lower arm mounting layer **231B**, both of the switching elements **31** and the protective element **32** are aligned in the second direction **x2**. As shown in FIG. 15, the third lower arm mounting layer **231B** is electrically connected, via wires **41**, to the front surface electrodes **311** of the switching elements **31** and the anode electrodes **321** of the protective elements **32** that are electrically bonded to the third upper arm mounting layer **231A**.

(90) As shown in FIGS. 10 and 18, the third electroconductive layer **232** is electrically connected, via wires **41**, to the front surface electrodes **311** of the switching elements **31** and the anode electrodes **321** of the protective elements **32** that are electrically bonded to the third lower arm mounting layer **231B**. The third electroconductive layer **232** is offset toward the other end of the third substrate **11C** (lower side in FIG. 10) in the first direction **x1**. The third electroconductive layer **232** is in the form of a strip extending along the second direction **x2**.

(91) As shown in FIGS. 15 and 18, the third gate layer **233** is electrically connected, via first gate wires **421**, to the gate electrodes **313** of the switching elements **31** electrically bonded to the third mounting layer **231**. The third gate layer **233** is in the form of a strip extending along the second direction **x2** and faces the switching elements **31** as viewed in the thickness direction **z**. The third gate layer **233** includes a third upper arm gate layer **233A** and a third lower arm gate layer **233B**.

(92) As shown in FIG. 10, the third upper arm gate layer **233A** is located between the third upper arm mounting layer **231A** and the case **70** in the first direction **x1**. As viewed in the thickness direction **z**, the third upper arm gate layer **233A** faces the switching elements **31** electrically bonded to the third upper arm mounting layer **231A**. As shown in FIG. 15, the third upper arm gate layer **233A** is electrically connected, via first gate wires **421**, to the gate electrodes **313** of the switching elements **31** electrically bonded to the third upper arm mounting layer **231A**.

(93) As shown in FIG. 10, the third lower arm gate layer **233B** is located between the third lower arm mounting layer **231B** and the third electroconductive layer **232** in the first direction **x1**. The third lower arm gate layer **233B** is in the form of an L-shaped strip with a part extending in the first direction **x1** and a part extending in the second direction **x2**. As viewed in the thickness direction **z**, the part extending in the second direction **x2** faces the switching elements **31** electrically bonded to the third lower arm mounting layer **231B**. As shown in FIG. 18, the third lower arm gate layer **233B** is electrically connected, via first gate wires **421**, to the gate electrodes **313** of the switching elements **31** electrically bonded to the third lower arm mounting layer **231B**.

(94) As shown in FIGS. 15 and 18, the third detection layer **234** is electrically connected, via first detection wires **431**, to the front surface electrodes **311** of the switching elements **31** electrically bonded to the third mounting layer **231**. The third detection layer **234** is in the form of a strip extending along the second direction **x2** and faces the switching elements **31** as viewed in the thickness direction **z**. The third detection layer **234** includes a third upper arm detection layer **234A** and a third lower arm detection layer **234B**.

(95) As shown in FIG. 10, the third upper arm detection layer 234A is located between the third upper arm mounting layer 231A and the third upper arm gate layer 233A in the first direction x1. As viewed in the thickness direction z, the third upper arm detection layer 234A faces the switching elements 31 electrically bonded to the third upper arm mounting layer 231A. As shown in FIG. 15, the third upper arm detection layer 234A is electrically connected, via first detection wires 431, to the front surface electrodes 311 of the switching elements 31 electrically bonded to the third upper arm mounting layer 231A.

(96) As shown in FIG. 10, the third lower arm detection layer 234B is located between the third lower arm mounting layer 231B and the third lower arm gate layer 233B in the first direction x1. As viewed in the thickness direction z, the third lower arm detection layer 234B faces the switching elements 31 electrically bonded to the third lower arm mounting layer 231B. As shown in FIG. 18, the third lower arm detection layer 234B is electrically connected, via first detection wires 431, to the front surface electrodes 311 of the switching elements 31 electrically bonded to the third lower arm mounting layer 231B.

(97) The first upper arm mounting layer 211A, the second upper arm mounting layer 221A and the third upper arm mounting layer 231A respectively correspond to sections of the “upper arm mounting layer” as set forth in the appended claims of the present disclosure. The first lower arm mounting layer 211B, the second lower arm mounting layer 221B and the third lower arm mounting layer 231B respectively correspond to sections of the “lower arm mounting layer” as set forth in the appended claims of the present disclosure.

(98) As shown in FIGS. 2 and 3, the power supply terminal 24 is an element of an external connection terminal provided in the semiconductor device A10. As described before, the power supply terminal 24 includes a first power supply terminal 24A and a second power supply terminal 24B. The power supply terminal 24 is supported on the case 70 and connected to a DC power supply arranged outside the semiconductor device A10. The power supply terminal 24 is made of a thin metal plate such as a copper plate. The surface of the thin metal plate may be plated with nickel (Ni). The first power supply terminal 24A is the positive electrode (P-terminal) of the semiconductor device A10. The second power supply terminal 24B is the negative electrode (N-terminal) of the semiconductor device A10. The first power supply terminal 24A and the second power supply terminal 24B are spaced apart from each other in the first direction x1. The first power supply terminal 24A and the second power supply terminal 24B have the same shape.

(99) As shown in FIG. 11, the power supply terminal 24 is bent into a hook shape as viewed in the first direction x1. The power supply terminal 24 is formed with a coupling hole 241 penetrating the terminal in the thickness direction z at a portion exposed outside the semiconductor device A10 and extending perpendicularly to the thickness direction z. Into the coupling hole 241, a fastening member such as a bolt is inserted. As shown in FIG. 8, a connection member 242 with electrical conductivity is connected to a portion of the power supply terminal 24 that is located inside the case 70 and that extends perpendicularly to the thickness direction z. For example, the connection member 242 includes a plurality of wires made of aluminum (Al). The connection member 242 connected to the first power supply terminal 24A is connected at its other end to the power supply pad 211C of the first upper arm mounting layer 211A. Thus, with this connection member 242, the first power supply terminal 24A is electrically connected to the first upper arm mounting layer 211A. The connection member 242 connected to the second power supply terminal 24B is connected, at its other end, to the second power supply pad 212A of the first electroconductive layer 212. Thus, with this connection member 242, the second power supply terminal 24B is electrically connected to the first electroconductive layer 212.

(100) As shown in FIGS. 2 and 3, the output terminal 25 is an element of an external connection terminal provided in the semiconductor device A10. The output terminal 25 is divided into two, namely a first output terminal 25A and a second output terminal 25B. Note that the output terminal 25 may be configured as a single unit that is not divided into multiple parts. The output terminal 25

is supported on the case **70** and connected to a driving target such as a motor arranged outside the semiconductor device **A10**. The output terminal **25** is located opposite to the power supply terminal **24** across the substrate **11** in the second direction x_2 . The output terminal **25** is made of the same thin metal film as the power supply terminal **24**. The surface of the thin metal plate may be plated with nickel. The first output terminal **25A** and the second output terminal **25B** are connected in parallel to the second lower arm mounting layer **221B**. The first output terminal **25A** and the second output terminal **25B** are each connected to a driving target of the semiconductor device **A10** that is externally arranged. In the second direction x_2 , the first output terminal **25A** faces the first power supply terminal **24A**, whereas the second output terminal **25B** faces the second power supply terminal **24B**. The first output terminal **25A** and the second output terminal **25B** are spaced apart from each other in the first direction x_1 . The first output terminal **25A** and the second output terminal **25B** have the same shape.

(101) As shown in FIG. **11**, the output terminal **25** is bent into a hook shape as viewed in the first direction x_1 . The output terminal **25** is formed with a coupling hole **251** penetrating the terminal in the thickness direction z at a portion exposed outside the semiconductor device **A10** and extending perpendicularly to the thickness direction z . Into the coupling hole **251**, a fastening member such as a bolt is inserted. As shown in FIG. **9**, a connection member **252** with electrical conductivity is connected to a portion of the output terminal **25** that is located inside the case and that extends perpendicularly to the thickness direction z . For example, the connection member **252** includes a plurality of wires made of aluminum. The connection member **252** connected to the output terminal **25** is connected at its other end to the output pad **221C** of the second lower arm mounting layer **221B** disposed on the second substrate **11B**. Thus, with the connection member **25**, the output terminal **25** is electrically connected to the second lower arm mounting layer **221B**.

(102) As shown in FIG. **10**, the connecting electroconductive member **261** connects the first mounting layer **211** and the third mounting layer **231** to each other, and also connects the second mounting layer **221** and the third mounting layer **231** to each other. Thus, the first mounting layer **211**, the second mounting layer **221** and the third mounting layer **231** are electrically connected to each other via the connecting electroconductive member **261**. Further, as shown in FIG. **10**, the connecting electroconductive member **261** connects the first electroconductive layer **212** and the third electroconductive layer **232** to each other, and also connects the second electroconductive layer **222** and the third electroconductive layer **232** to each other. Thus, the first electroconductive layer **212**, the second electroconductive layer **222** and the third electroconductive layer **232** are electrically connected to each other via the connecting electroconductive member **261**. For example, the connecting electroconductive member **261** includes a plurality of wires made of aluminum.

(103) As shown in FIG. **10**, the connecting electroconductive member **261** includes a first part **261A**, a second part **261B** and a third part **261C**. All of the first part **261A**, the second part **261B** and the third part **261C** extend in the second direction x_2 . The first part **261A** connects the first upper arm mounting layer **211A** and the third upper arm mounting layer **231A** to each other, and also connects the second upper arm mounting layer **221A** and the third upper arm mounting layer **231A** to each other. Thus, the first upper arm mounting layer **211A** and the second upper arm mounting layer **221A** are electrically connected to each other via the first part **261A**. The second part **261B** connects the first lower arm mounting layer **211B** and the third lower arm mounting layer **231B** to each other, and also connects the second lower arm mounting layer **221B** and the third lower arm mounting layer **231B** to each other. Thus, the first lower arm mounting layer **211B** and the second lower arm mounting layer **221B** are electrically connected to each other via the second part **261B**. The third part **261C** connects the first electroconductive layer **212** and the third electroconductive layer **232** to each other, and also connects the second electroconductive layer **222** and the third electroconductive layer **232** to each other. Thus, the first electroconductive layer **212** and the second electroconductive layer **222** are electrically connected to each other via the third

part **261C**.

(104) As shown in FIG. **10**, first electroconductive members **262** connect the first gate layer **213** and the third gate layer **233** to each other, and also connect the second gate layer **223** and the third gate layer **233** to each other. Thus, the first gate layer **213**, the second gate layer **223** and the third gate layer **233** are electrically connected to each other via the first electroconductive members **262**. For example, the first electroconductive members **262** are wires made of aluminum. All of the first electroconductive members **262** extend in the second direction x_2 and may consist of four first electroconductive members **262**. The first one of the first electroconductive members **262** connects the first upper arm gate layer **213A** and the third upper arm gate layer **233A**. The second one of the first electroconductive members **262** connects the second upper arm gate layer **223A** and the third upper arm gate layer **233A**. The third one of the first electroconductive members **262** connects the first lower arm gate layer **213B** and the third lower arm gate layer **233B**. The fourth one of the first electroconductive members **262** connects the second lower arm gate layer **223B** and the third lower arm gate layer **233B**.

(105) As shown in FIG. **10**, second electroconductive members **263** connect the first detection layer **214** and the third detection layer **234** to each other, and also connect the second detection layer **224** and the third detection layer **234** to each other. Thus, the first detection layer **214**, the second detection layer **224** and the third detection layer **234** are electrically connected to each other via the second electroconductive members **263**. For example, the second electroconductive members **263** are wires made of aluminum. All of the second electroconductive members **263** extend in the second direction x_2 and may consist of four second electroconductive members **263**. The first one of the second electroconductive members **263** connects the first upper arm detection layer **214A** and the third upper arm detection layer **234A**. The second one of the second electroconductive members **263** connects the second upper arm detection layer **224A** and the third upper arm detection layer **234A**. The third one of the second electroconductive members **263** connects the first lower arm detection layer **214B** and the third lower arm detection layer **234B**. The fourth one of the second electroconductive members **263** connects the second lower arm detection layer **224B** and the third lower arm detection layer **234B**.

(106) As shown in FIGS. **2-4**, a gate terminal **27** is an element of an external connection terminal provided in the semiconductor device **A10**. The gate terminal **27** is connected to an externally arranged driving circuit (e.g. gate driver) for the semiconductor device **A10**. The gate terminal **27** is disposed to face the substrate **11** as viewed in the thickness direction z and supported on the case **70**. The gate terminal **27** projects in the same direction in which the front surface **111** of the substrate **11** faces (along the thickness direction z). For example, the gate terminal **27** is in the form of a metal rod made of copper. The surface of the metal rod is plated with tin (Sn). Nickel plating may be provided between the surface of the metal rod and the tin plating. As shown in FIG. **12**, the gate terminal **27** is bent into a hook shape at its end closer to the substrate **11** in the thickness direction z , thereby having a portion extending along the first direction x_1 . The gate terminal **27** includes a first gate terminal **27A** and a second gate terminal **27B**. Paired second gate wires **422** are connected to the first gate terminal **27A** and the second gate terminal **27B**. For example, the paired second gate wires **422** are made of aluminum.

(107) As shown in FIG. **10**, the first gate terminal **27A** is disposed close to the second upper arm gate layer **223A** to face the second substrate **11B** as viewed in the thickness direction z . The second gate wire **422** connected at one end to the first gate terminal **27A** is connected at the other end to the third upper arm gate layer **233A**. Thus, the first gate terminal **27A** is electrically connected to the gate electrodes **313** of the switching elements **31** electrically bonded to the first upper arm mounting layer **211A**, the second upper arm mounting layer **221A** and the third upper arm mounting layer **231A**.

(108) As shown in FIG. **10**, the second gate terminal **27B** is disposed close to the third lower arm gate layer **233B** to face the third substrate **11C** as viewed in the thickness direction z . The second

gate wire **422** connected at one end to the second gate terminal **27B** is connected at the other end to the third lower arm gate layer **233B**. Thus, the second gate terminal **27B** is electrically connected to the gate electrodes **313** of the switching elements **31** electrically bonded to the first lower arm mounting layer **211B**, the second lower arm mounting layer **221B** and the third lower arm mounting layer **231B**.

(109) As shown in FIGS. **2-4**, a device current detection terminal **281** is an element of an external connection terminal provided in the semiconductor device **A10**. The device current detection terminal **281** is connected to an externally arranged control circuit for the semiconductor device **A10**. The device current detection terminal **281** is disposed to face the substrate **11** and supported on the case **70**. The device current detection terminal **281** projects in the same direction in which the gate terminal **27** projects along the thickness direction **z**. The device current detection terminal **281** is made of a metal rod of the same material as the gate terminal **27**. The device current detection terminal **281** has the same shape as the gate terminal **27**. Thus, the device current detection terminal **281** is bent into a hook shape at its end closer to the substrate **11** in the thickness direction **z**, thereby having a portion extending along the first direction **x1**. The device current detection terminal **281** includes a first detection terminal **281A** and a second detection terminal **281B**. Paired second detection wires **432** are connected to the first detection terminal **281A** and the second detection terminal **281B**. For example, the paired second detection wires **432** are made of aluminum.

(110) As shown in FIG. **10**, the first detection terminal **281A** is disposed close to the second upper arm detection layer **224A** to face the second substrate **11B** as viewed in the thickness direction **z**, and is also close to the first gate terminal **27A**. The second detection wire **432** connected at one end to the first detection terminal **281A** is connected at the other end to the second upper arm detection layer **224A**. Thus, the first detection terminal **281A** is electrically connected to the front surface electrodes **311** of the switching elements **31** electrically bonded to the first upper arm mounting layer **211A**, the second upper arm mounting layer **221A** and the third upper arm mounting layer **231A**.

(111) As shown in FIG. **10**, the second detection terminal **281B** is disposed close to the first lower arm detection layer **214B** to face the first substrate **11A** as viewed in the thickness direction **z**, and is also close to the second gate terminal **27B**. The second detection wire **432** connected at one end to the second detection terminal **281B** is connected at the other end to the first lower arm detection layer **214B**. Thus, the second detection terminal **281B** is electrically connected to the front surface electrodes **311** of the switching elements **31** electrically bonded to the first lower arm mounting layer **211B**, the second lower arm mounting layer **221B** and the third lower arm mounting layer **231B**.

(112) As shown in FIGS. **2-4** and **9**, a power supply current detection terminal **281** is an element of an external connection terminal provided in the semiconductor device **A10**. The power supply current detection terminal **282** is connected to an externally arranged control circuit for the semiconductor device **A10** and supported on the case **70**. The power supply current detection terminal **282** projects in the same direction in which the gate terminal **27** projects along the thickness direction **z**. The power supply current detection terminal **28** is made of a metal rod of the same material as the gate terminal **27**. The power supply current detection terminal **282** is located at the same position as the first gate terminal **27A** and the first detection terminal **281A** in the first direction **x1** and spaced apart from the first detection terminal **281A** toward the first output terminal **25A** in the second direction **x2**. The power supply current detection terminal **282** is disposed close to the second upper arm mounting layer **221A** to face the second substrate **11B** in the first direction **x1**. The power supply current detection terminal **282** has the same shape as the gate terminal **27**. Thus, the power supply current detection terminal **282** is bent into a hook shape at its end closer to the second substrate **11B** in the thickness direction **z**, thereby having a portion extending along the first direction **x1**. To this end of the power supply current detection terminal, one end of a power

supply current detection wire **44** is connected. The other end of the power supply current detection wire **44** is connected to the second upper arm mounting layer **221A**. Thus, the power supply current detection terminal **282** is electrically connected to the first upper arm mounting layer **211A**, the second upper arm mounting layer **221A** and the third upper arm mounting layer **231A**. For example, the power supply current detection wire **44** is made of aluminum.

(113) As shown in FIGS. **2-4** and **8**, a pair of thermistor terminals **29** are an element of an external connection terminal provided in the semiconductor device **A10**. The paired thermistor terminals **29** are connected to an externally arranged control circuit for the semiconductor device **A10** and supported on the case **70**. The paired thermistor terminal **29** project in the same direction in which the gate terminal **27** projects along the thickness direction **z**. The paired thermistor terminals **29** are made of a metal rod of the same material as the gate terminal **27**. The paired thermistor terminals **29** are located at the same position as the first gate terminal **27A** and the first detection terminal **281A** in the first direction **x1** and spaced apart from the first gate terminal **27A** toward the first power supply terminal **24A** in the second direction **x2**. The paired thermistor terminal **29** are disposed close to the thermistor mounting layer **215** to face the first substrate **11A** in the first direction **x1**. The paired thermistor terminals **29** have the same shape as the gate terminal **27**. Thus, each of the paired thermistor terminals is bent into a hook shape at its end closer to the first substrate **11A** in the thickness direction **z**, thereby having a portion extending along the first direction **x1**. To this end of each of the paired thermistor terminals **29**, one end of a corresponding one of a paired thermistor wires **45** is connected. The other ends of the paired thermistor wires **45** are connected to paired sections of the thermistor mounting layer **215**. Thus, the thermistor terminals **29** are electrically connected to the thermistor **33**. For example, the paired thermistor wires **45** are made of aluminum.

(114) As shown in FIG. **3**, the switching elements **31** are semiconductor elements electrically bonded to and aligned, in the second direction **x2**, on each of the first mounting layer **211**, the second mounting layer **221** and the third mounting layer **231**. The switching elements **31** are rectangular (square in the semiconductor device **A10**) as viewed in the thickness direction **z**. The switching elements **31** are MOSFETs (Metal-Oxide-Semiconductor Field-Effect Transistors) made of a semiconductor material mainly composed of silicon carbide (SiC). Note that the switching elements **31** are not limited to MOSFETs and may be IGBTs (Insulated Gate Bipolar Transistors). In the semiconductor device **A10**, it is assumed that the switching elements **31** are n-channel MOSFETs made of a semiconductor material mainly composed of silicon carbide. In the semiconductor device **A10**, the switching elements **31** are 400 μm or less and preferably 150 μm or less in thickness. The breakdown voltage of the switching elements **31** is 1,200 V or more.

(115) As shown in FIGS. **15-20**, each of the switching elements **31** has a front surface **31A**, a back surface **31B**, a side surface **31C**, a front surface electrode **311**, a back surface electrode **312**, a gate electrode **313** and an insulating film **314**. The front surface **31A**, the back surface **31B** and the side surface **31C** correspond to the “first element front surface”, the “first element back surface” and the “first element side surface”, respectively, as set forth in the appended claims of the present disclosure. The front surface **31A** faces in the same direction in which the front surface **111** of the substrate **11** faces along the thickness direction **z**. The back surface **31B** faces in the opposite direction of the front surface **31A**. The switching elements **31** are electrically bonded to the first mounting layer **211**, the second mounting layer **221** and the third mounting layer **231**, with the back surfaces **31B** facing the front surface **111**. The side surface **31C** is connected to both of the front surface **31A** and the back surface **31B**. The side surface **31C** includes a plurality of sections (four sections in the semiconductor device **A10**) each facing in the first direction **x1** or the second direction **x2**.

(116) As shown in FIGS. **15-20**, the front surface electrode **311** is provided on the front surface **31A**. A source current flows through the front surface electrode **311**. The front surface electrode **311** has a pair of first pads **311A** and a pair of second pads **311B**. The paired first pads **311A** are

sections of a front surface electrode **311** that are spaced apart from each other in the second direction **x**, so are the paired second pads **311B**. In each of the switching elements **31** electrically bonded to the first mounting layer **211**, the second pads **311B** are located opposite to the first lower arm mounting layer **211B** or the first electroconductive layer **212** across the paired first pads **311A** in the first direction **x1**. The above-described positional relationship between the paired first pads **311A** and the paired second pads **311B** also applies to the switching elements **31** electrically bonded to the second mounting layer **221** or the third mounting layer **231**.

(117) As shown in FIG. **15**, in each of the switching elements **31** electrically bonded to the first upper arm mounting layer **211A**, the second upper arm mounting layer **221A** or the third upper arm mounting layer **231A**, one end of the first detection wire **431** is connected to one of the paired second pads **311B**. The other end of the first detection wire **431** is connected to the first upper arm detection layer **214A**, the second upper arm detection layer **224A** or the third upper arm detection layer **234A**. As shown in FIG. **18**, in each of the switching elements **31** electrically bonded to the first lower arm mounting layer **211B**, the second lower arm mounting layer **221B** or the third lower arm mounting layer **231B**, one end of the first detection wire **431** is connected to one of the paired first pads **311A**. The other end of the first detection wire **431** is connected to the first lower arm detection layer **214B**, the second lower arm detection layer **224B** or the third lower arm detection layer **234B**. In this way, the front surface electrodes **311** are electrically connected to the first detection layer **214**, the second detection layer **224** or the third detection layer **234** via the first detection wires **431**. For example, the first detection wires **431** are made of gold (Au).

(118) As shown in FIGS. **16-20** (excluding FIG. **18**), the back surface electrode **312** is provided on the entirety of the back surface **31B**. A drain current flows through the back surface electrode **312**. The back surface electrode **312** is electrically bonded to one of the first mounting layer **211**, the second mounting layer **221** and the third mounting layer **231** via a first bonding layer **391**. The first bonding layer **391** is electrically conductive. The first bonding layer **391** is sandwiched between the back surface electrode **312** and the first mounting layer **211**, the second mounting layer **221** or the third mounting layer **231**. For example, the first bonding layer **391** is made of lead-free solder mainly composed of tin. The first bonding layer **391** electrically connects each of the back surface electrodes **312** to one of the first mounting layer **211**, the second mounting layer **221** and the third mounting layer **231**.

(119) As shown in FIGS. **15** and **18**, the gate electrode **313** is provided on the front surface **31A**. A gate voltage for driving each of the switching elements **31** is applied to the gate electrode **313**. As shown in FIG. **15**, in each of the switching elements **31** electrically bonded to the first upper arm mounting layer **211A**, the second upper arm mounting layer **221A** or the third upper arm mounting layer **231A**, the gate electrode **313** is close to the paired second pads **311B** of the front surface electrode **311**. One end of a first gate wire **421**, which is connected at its other end to the first upper arm gate layer **213A**, the second upper arm gate layer **223A** or the third upper arm gate layer **233A**, is connected to the gate electrode **313**. As shown in FIG. **18**, in each of the switching elements **31** electrically bonded to the first lower arm mounting layer **211B**, the second lower arm mounting layer **221B** or the third lower arm mounting layer **231B**, the gate electrode **313** is close to the paired first pads **311A** of the front surface electrode **311**. One end of a first gate wire **421**, which is connected at its other end to the first lower arm gate layer **213B**, the second lower arm gate layer **223B** or the third lower arm gate layer **233B**, is connected to the gate electrode **313**. In this way, the gate electrodes **313** are electrically connected to the first gate layer **213**, the second gate layer **223** or the third gate layer **233** via the first gate wires **421**. For example, the first gate wires **421** are made of gold.

(120) As shown in FIGS. **15-20**, the insulating film **314** is provided on the front surface **31A**. The insulating film **314** is electrically insulating. As viewed in the thickness direction **z**, the insulating film **314** surrounds the front surface electrode **311**. The insulating film **314** may be formed by laminating a silicon dioxide (SiO₂) layer, a silicon nitride (Si₃N₄) layer, and a

polybenzoxazole (PBO) layer on the front surface **31A** in the mentioned order. For the insulating film **314**, a polyimide layer may be used instead of the polybenzoxazole layer. In FIGS. **15-20**, the length from the edge **314A** of the insulating film **314** to the front surface electrode **311** as viewed in the thickness direction z is indicated as a gap G_p in each of the switching elements **31**. The gap G_p is a length along the first direction x_1 or the second direction x_2 . The edge **314A** is rectangular (square in the semiconductor device **A10**) as viewed in the thickness direction z . As viewed in the thickness direction z , the ratio of the length of the gap G_p to the length of one side of the edge **314** (shorter side when the edge **314** is rectangular) is set to 5% to 25%. The longer the gap G_p is, the higher the dielectric breakdown voltage of the switching element **31** is.

(121) As shown in FIG. **3**, the protective elements **32** are semiconductor elements electrically bonded to and aligned in the second direction x_2 on each of the first mounting layer **211**, the second mounting layer **221** and the third mounting layer **231**. The protective elements **32** are rectangular as viewed in the thickness direction z . The protective elements **32** are arranged to be electrically connected to the switching elements **31**, respectively. The protective elements **32** are electrically connected to both of the front surface electrodes **311** and the back surface electrodes **312** of the switching elements **31**. Thus, each of the switching elements **31** and a corresponding one of the protective elements **32** form a parallel circuit. For example, the protective elements **32** are schottky-barrier diodes made by using a semiconductor material mainly composed of silicon carbide. In the semiconductor device **A10**, the protective elements **32** are 400 μm or less and preferably 150 μm or less in thickness. The breakdown voltage of the protective elements **32** is 1,200 V or more.

(122) As shown in FIGS. **15-20**, each of the protective elements **32** has a front surface **32A**, a back surface **32B**, a side surface **32C**, an anode electrode **321**, a cathode electrode **322** and an insulating film **323**. The front surface **32A** corresponds to the “second element front surface” as set forth in the appended claims of the present disclosure. The front surface **32A** faces in the same direction in which the front surface **111** of the substrate **11** faces along the thickness direction z . The back surface **32B** faces in the opposite direction of the front surface **32A**. The protective elements **32** are electrically bonded to the first mounting layer **211**, the second mounting layer **221** and the third mounting layer **231**, with the back surfaces **32B** facing the front surface **111**. The side surface **32C** is connected to both of the front surface **32A** and the back surface **32B**. The side surface **32C** includes a plurality of sections (four sections in the semiconductor device **A10**) each facing in the first direction x_1 or the second direction x_2 .

(123) As shown in FIGS. **15-20**, the anode electrode **321** is provided on the front surface **32A**. The anode electrode **321** is electrically connected to the front surface electrode **311** of the switching element **31** with which that protective element **32** is associated.

(124) As shown in FIGS. **16** and **19**, the cathode electrode **322** is provided on the entirety of the back surface **32B**. The cathode electrode **322** is electrically bonded to one of the first mounting layer **211**, the second mounting layer **221** and the third mounting layer **231** via a second bonding layer **392**. The second bonding layer **392** is electrically conductive. The second bonding layer **392** is sandwiched between the cathode electrode **322** and the first mounting layer **211**, the second mounting layer **221** or the third mounting layer **231**. The second bonding layer **392** is made of the same material as that for the first bonding layer **391**. With the second bonding layer **392**, the cathode electrode **322** is electrically connected, via the first mounting layer **211**, the second mounting layer **221** or the third mounting layer **231**, to the back surface electrode **312** of the switching element **31** with which the protective element **32** of that cathode electrode **322** is associated.

(125) As shown in FIGS. **16** and **19**, the insulating film **323** is provided on the front surface **32A**. The insulating film **323** is electrically insulating. As shown in FIGS. **15** and **18**, the insulating film **323** surrounds the anode electrode **321** as viewed in the thickness direction z . The insulating film **323** may be formed by laminating a silicon dioxide layer, a silicon nitride layer, and a

polybenzoxazole layer on the front surface **31A** in the mentioned order. For the insulating film **323**, a polyimide layer may be used instead of the polybenzoxazole layer.

(126) As shown in FIGS. **3** and **8**, the thermistor **33** is an element electrically bonded to the thermistor mounting layer **215**. For example, the thermistor **33** is an NTC (Negative Temperature Coefficient) thermistor. An NTC thermistor has the property of decreasing the resistance with increasing temperature. The thermistor **33** is used as a temperature detection sensor of the semiconductor device **A10**.

(127) As shown in FIGS. **15-17**, wires **41** are connected to the front surface electrodes **311** of switching elements **31** and the first lower arm mounting layer **211B**, the second lower arm mounting layer **221B** or the third lower arm mounting layer **231B**. As shown in FIGS. **18-20**, wires **41** are connected to the front surface electrodes **311** of switching elements **31** and the first electroconductive layer **212**, the second electroconductive layer **222** or the third electroconductive layer **232**. For example, the wires **41** are made of aluminum. The wires **41** are larger in diameter than the first gate wires **421** and the first detection wires **431**.

(128) As shown in FIGS. **15-17**, in the switching elements **31** electrically bonded to the first upper arm mounting layer **211A**, wires **41** are connected to the front surface electrodes **311** and the first lower arm mounting layer **211B**. As shown in FIGS. **18-20**, in the switching elements **31** electrically bonded to the first lower arm mounting layer **211B**, wires **41** are connected to the front surface electrodes **311** and the first electroconductive layer **212**. Thus, the front surface electrodes **311** of the switching elements **31** electrically bonded to the first mounting layer **211** are electrically connected to the first lower arm mounting layer **211B** or the first electroconductive layer **212**.

(129) As shown in FIGS. **15-17**, in the switching elements **31** electrically bonded to the second upper arm mounting layer **221A**, wires **41** are connected to the front surface electrodes **311** and the second lower arm mounting layer **221B**. As shown in FIGS. **18-20**, in the switching elements **31** electrically bonded to the second lower arm mounting layer **221B**, wires **41** are connected to the front surface electrodes **311** and the second electroconductive layer **222**. Thus, the front surface electrodes **311** of the switching elements **31** electrically bonded to the second mounting layer **221** are electrically connected to the second lower arm mounting layer **221B** or the second electroconductive layer **222**.

(130) As shown in FIGS. **15-17**, in the switching elements **31** electrically bonded to the third upper arm mounting layer **231A**, wires are connected to the front surface electrodes **311** and the third lower arm mounting layer **231B**. As shown in FIGS. **18-20**, in the switching elements **31** electrically bonded to the third lower arm mounting layer **231B**, wires **41** are connected to the front surface electrodes **311** and the third electroconductive layer **232**. Thus, the front surface electrodes **311** of the switching elements **31** electrically bonded to the third mounting layer **231** are electrically connected to the third lower arm mounting layer **231B** or the third electroconductive layer **232**.

(131) As shown in FIGS. **15-20**, the wires **41** extend in the first direction **x1**. Each of the wires **41** has a first bonding portion **411**. The first bonding portions **411** are held in contact with the front surface electrodes **311** of the switching elements **31**. The wires **41** for each of the switching elements **31** include a pair of inner wires **41A** and a pair of outer wires **41B**. The paired inner wires **41A** are flanked by the paired outer wires **41B** in the second direction **x2**.

(132) With reference to FIGS. **15-17**, description is given below of the configuration of the wires **41** for each of the switching elements **31** electrically bonded to the first upper arm mounting layer **211A**, the second upper arm mounting layer **221A** or the third upper arm mounting layer **231A**. As shown in FIG. **17**, the first bonding portions **411** of the paired inner wires **41A** are held in contact with the paired first pads **311A** of the front surface electrode **311**. As shown in FIG. **16**, the first bonding portions **411** of the paired outer wires **41B** are held in contact with both of the paired first pads **311A** and the paired second pads **311B** of the front surface electrode **311**. As shown in FIGS. **15** and **16**, each of the first bonding portions **411** of the paired outer wires **41B** has a first connect

portion **411A**, a second connect portion **411B** and a joint portion **411C**. The first connect portion **411A** is held in contact with a first pad **311A**. The second connect portion **411B** is held in contact with a second pad **311B**. The joint portion **411C** is sandwiched between the first connect portion **411A** and the second connect portion **411B** in the first direction **x1**. The joint portion **411C** projects in the same direction in which the front surface **31A** of the switching element **31** faces along the thickness direction **z**.

(133) As shown in FIGS. **15** and **16**, each of the wires **41** for each of the switching elements **31** electrically bonded to the first upper arm mounting layer **211A**, the second upper arm mounting layer **221A** or the third upper arm mounting layer **231A** has a second bonding portion **412**. The second bonding portion **412** is held in contact with the anode electrode **321** of the protective element **32**. Thus, the anode electrodes **321** of the protective elements **32** electrically bonded to the first upper arm mounting layer **211A** are electrically connected to both of the front surface electrodes **311** of the corresponding switching elements **31** and the first lower arm mounting layer **211B**. The anode electrodes **321** of the protective elements **32** electrically bonded to the second upper arm mounting layer **221A** are electrically connected to both of the front surface electrodes **311** of the corresponding switching elements **31** and the second lower arm mounting layer **221B**. The anode electrodes **321** of the protective elements **32** electrically bonded to the third upper arm mounting layer **231A** are electrically connected to both of the front surface electrodes **311** of the corresponding switching elements **31** and the third lower arm mounting layer **231B**.

(134) With reference to FIGS. **18-20**, description is given below of the configuration of the wires **41** for each of the switching elements **31** electrically bonded to the first lower arm mounting layer **211B**, the second lower arm mounting layer **221B** or the third lower arm mounting layer **231B**. As shown in FIG. **20**, the first bonding portions **411** of the paired inner wires **41A** are held in contact with the paired first pads **311A** of the front surface electrode **311**. As shown in FIG. **19**, the first bonding portions **411** of the paired outer wires **41B** are held in contact with both of the paired first pads **311A** and the paired second pads **311B** of the front surface electrode **311**. As shown in FIGS. **18** and **19**, each of the first bonding portions **411** of the paired outer wires **41B** has a first connect portion **411A**, a second connect portion **411B** and a joint portion **411C**. The first connect portion **411A** is held in contact with a first pad **311A**. The second connect portion **411B** is held in contact with a second pad **311B**. The joint portion **411C** is sandwiched between the first connect portion **411A** and the second connect portion **411B** in the first direction **x1**. The joint portion **411C** projects in the same direction in which the front surface **31A** of the switching elements **31** faces along the thickness direction **z**.

(135) As shown in FIGS. **18** and **19**, each of the paired outer wires **41B** for each of the switching elements **31** electrically bonded to the first lower arm mounting layer **211B**, the second lower arm mounting layer **221B** or the third lower arm mounting layer **231B** has a second bonding portion **412**. The second bonding portion **412** is held in contact with the anode electrode **321** of the protective element **32**. Thus, the anode electrodes **321** of the protective elements **32** electrically bonded to the first lower arm mounting layer **211B** are electrically connected to both of the front surface electrodes **311** of the corresponding switching elements **31** and the first electroconductive layer **212**. The anode electrodes **321** of the protective elements **32** electrically bonded to the second lower arm mounting layer **221B** are electrically connected to both of the front surface electrodes **311** of the corresponding switching elements **31** and the second electroconductive layer **222**. The anode electrodes **321** of the protective elements **32** electrically bonded to the third lower arm mounting layer **231B** are electrically connected to both of the front surface electrodes **311** of the corresponding switching elements **31** and the third electroconductive layer **232**.

(136) As shown in FIG. **18**, a pair of auxiliary wires **46** are connected to the anode electrode **321** of each of the protective elements **32** electrically bonded to the first lower arm mounting layer **211B**, the second lower arm mounting layer **221B** or the third lower arm mounting layer **231B**. The auxiliary wires **46** are connected at the other ends thereof to paired second pads **311B** of the front

surface electrode **311** of the switching element **31** with which the protective element **32** is associated. The auxiliary wires **46** are located between paired outer wires **41B** in the second direction **x2**. The auxiliary wires **46** are made of the same material as that for the wires **41**. The diameter of the auxiliary wires **46** are equal to that of the wires **41**.

(137) As shown in FIGS. **15-20**, the moisture-resistant layer **51** covers the side surface **31C** of switching elements **31**. As the material for the moisture-resistant layer **51**, an electrically insulating material having a high resistance to temperature cycling and a lower moisture permeability than the sealing resin **52** (silicone gel in the semiconductor device **A10**) is selected. As such an electrically insulating material for the moisture-resistant layer **51**, polyimide and silicone gel are selected. The ratio of content by weight of polyimide to silicone gel in the moisture-resistant layer **51** is 1.5:1 to 7.0:1. That is, in the moisture-resistant layer **51**, the weight of polyimide is larger than that of silicone gel. In the moisture-resistant layer **51**, polyimide molecules and silicone gel molecules are mixed. Preferably, polyimide molecules and silicone gel molecules are uniformly dispersed throughout the moisture-resistant layer **51**. This effectively prevents the moisture-resistant layer **51** from cracking due to temperature cycling, so that the moisture-resistant layer **51** maintains the function of preventing moisture intrusion. Although the moisture-resistant layer **51** consisting of polyimide and silicone gel alone is explained for the semiconductor device **A10**, other materials may be added to these materials to form the moisture-resistant layer **51**. Further, although a mixture of polyimide and silicone gel is selected as the material for the moisture-resistant layer **51** in the semiconductor device **A10**, other materials having low moisture permeability may be selected. For example, the moisture-resistant layer **51** may be made of a mixture of polybenzoxazole and silicone gel.

(138) An example of a method for forming the moisture-resistant layer **51** of the semiconductor device **A10** is described below. A liquefied synthetic resin material containing polyimide, silicone gel and solvent is prepared. Note that the solvent is volatile. Then, the synthetic resin material is dropped with a dispenser onto the mounting layer (the first mounting layer **211**, the second mounting layer **221** and the third mounting layer **231**). As a result, the synthetic resin material spreads on the side surfaces **31C** of the switching elements **31**, so that the side surfaces **31C** are covered with the synthetic resin material. Finally, the synthetic resin material is heat-cured to obtain the moisture-resistant layer **51**. In this process, the solvent volatilizes. With this method, the moisture-resistant layer **51** covering the side surfaces **31C** of the switching elements **31** is easily formed.

(139) As shown in FIGS. **15-17**, the moisture-resistant layer **51** is held in contact with one of the first upper arm mounting layer **211A**, the second upper arm mounting layer **221A** or the third upper arm mounting layer **231A** and with the side surface **31C** of at least one of the switching elements **31**. In the thickness direction **z**, the moisture-resistant layer **51** extends to be spanned between the side surface **31C** and the first upper arm mounting layer **211A** or the second upper arm mounting layer **221A** or the third upper arm mounting layer **231A**, thereby crossing over the bonding layer **39** and the back surface electrode **312**.

(140) As shown in FIGS. **18-20**, the moisture-resistant layer **51** is held in contact with the first lower arm mounting layer **211B**, the second lower arm mounting layer **221B** or the third lower arm mounting layer **231B** and with the side surface **31C** of switching elements **31**. In the thickness direction **z**, the moisture-resistant layer **51** extends to be spanned between the side surface **31C** and the first lower arm mounting layer **211B** or the second lower arm mounting layer **221B** or the third lower arm mounting layer **231B**, thereby crossing over the bonding layer **39** and the back surface electrode **312**.

(141) Thus, the moisture-resistant layer **51** is held in contact with the first mounting layer **211**, the second mounting layer **221** or the third mounting layer **231** and with at least one of the side surfaces **31C**. In the thickness direction, the moisture-resistant layer **51** extends to be spanned between the first mounting layer **211**, the second mounting layer **221** or the third mounting layer

231 and the side surface **31C**.

(142) As shown in FIGS. **15**, **16**, **18** and **19**, the moisture-resistant layer **51** integrally covers the side surface **31C** of a switching element **31** and the side surface **32C** of the protective element **32** paired with that switching element **31** (the protective element **32** connected anti-parallel to the switching element **31**). In the example shown in these figures, the moisture-resistant layer **51** is provided correspondingly to the pairs of switching elements **31** and protective element **32**. That is, the moisture-resistant layer **51** is divided into a plurality of sections such that each section covers a pair of the side surface **31C** of a switching element **31** and the side surface **32C** of a protective element **32**. Unlike this configuration, the moisture-resistant layer **51** may be configured to integrally cover the side surfaces **31** of a plurality of switching elements **31** on each of the first mounting layer **211**, the second mounting layer **221** and the third mounting layer **231**.

(143) As shown in FIGS. **11** and **12**, the sealing resin **52** is housed in a region surrounded by the case **70** and the heat sink **61**. As shown in FIGS. **16**, **17**, **19** and **20**, the sealing resin **52** covers both of the switching elements **31** and the moisture-resistant layer **51**. The sealing resin **52** also covers the protective elements **32**. It is preferable that the sealing resin **52** is an electrically insulating synthetic resin with excellent heat resistance and adhesion. For example, the sealing resin **52** is silicone gel mainly composed of thermosetting organopolysiloxane. The sealing resin **52** is exposed to the atmosphere.

(144) As shown in FIGS. **11** and **12**, the heat sink **61** is bonded to the back surface **112** of the substrate **11**. In the semiconductor device **A10**, the heat sink **61** is bonded to each of the back surface **112** of the first substrate **11A**, the back surface **112** of the second substrate **11B** and the back surface **112** of the third substrate **11C** via a heat transfer layer **62** and a substrate bonding layer **69** (both described later). For example, the heat sink **61** is made of a metal plate such as a copper plate. The surface of the metal plate may be plated with nickel. As shown in FIGS. **7-9**, the heat sink **61** is provided with a plurality of support holes **611** at its four corners as viewed in the thickness direction **z**. Each of the support holes **611** penetrates the heat sink **61** in the thickness direction **z**. The support holes **611** are used to support the heat sink **61**, which is bonded to the substrate **11**, on the case **70**.

(145) As shown in FIGS. **11** and **12**, the heat transfer layer **62** is disposed on the back surface **112** of the substrate **11**. The heat transfer layer **62** is made of a metallic material such as a copper foil. The heat transfer layer **62** transfers the heat generated by driving the switching elements **31** to the heat sink **61**.

(146) The substrate bonding layer **69** is a bonding material interposed between the heat sink **61** and the heat transfer layer **62**, as shown in FIGS. **11** and **12**. In the semiconductor device **A10**, the substrate bonding layer **69** is made of lead-free solder mainly composed of tin. The substrate bonding layer **69** bonds the heat sink **61** to the substrate **11**.

(147) The case **70** is an electrically insulating member surrounding the substrate **11** as viewed in the thickness direction **z**, as shown in FIG. **3**. The case **70** is in the form of a frame. The case **70** is made of an electrically insulating synthetic resin with excellent heat resistance such as PPS (polyphenylene sulfide). The case **70** has a pair of side walls **71**, a pair of terminal seats **72**, mount parts **73**, a power supply terminal base **74** and an output terminal base **75**.

(148) As shown in FIGS. **2**, **3**, **5** and **6**, the paired side walls **71** are spaced apart from each other in the first direction **x1** and in the form of a groove. Each of the side walls **71** is arranged along both of the second direction **x2** and the thickness direction **z**, and one end of each side wall in the thickness direction **z** is held in contact with the heat sink **61**. Opposite ends of each side wall **71** in the second direction **x2** are connected to the paired terminal seats **72**. In one of the side walls **71**, the first gate terminal **27A**, the first detection terminal **281A**, the power supply current detection terminal **282** and the paired thermistor terminals **29** are disposed. In the other side wall **71**, the second gate terminal **27B** and the second detection terminal **281B** are disposed. As shown in FIGS. **8-10**, the ends of these terminals that are close to the substrate **11** in the thickness direction **z** are

supported on the side walls **71**.

(149) As shown in FIGS. **3**, **8** and **9**, the paired terminal seats **72** are spaced apart from each other in the second direction x_2 . Each of the terminal seats **72** is disposed along the second direction x_2 . To one of the terminal seats **72** is connected the power supply terminal base **74** that projects outward in the second direction x_2 , and a part of the power supply terminal **24** is supported on the terminal seat **72**. To the other one of the terminal seats **72** is connected the output terminal base **75** that projects outward in the second direction x_2 , and a part of the output terminal **25** is supported on the terminal seat **72**.

(150) As shown in FIGS. **2**, **8** and **9**, the mount parts **73** are provided at four corners of the case **70** as viewed in the thickness direction z . Each of the mount parts **73** is provided with a mount hole **731** penetrating the mount part **73** in the thickness direction z . The positions of the mount holes **731** correspond to the support holes **611** provided in the heat sink **61**. The heat sink **61** is supported on the case **70** by inserting fastening members such as pins into the mount holes **731** and the support holes **611**.

(151) As shown in FIGS. **2**, **5** and **8**, the power supply terminal base **74**, along with the terminal seat **72** connected thereto, supports the power supply terminal **24**. The power supply terminal base **74** includes a first terminal base **741** and a second terminal base **742**. The first terminal base **741** and the second terminal base **742** are spaced apart from each other in the first direction x_1 . On the first terminal base **741** is supported a part of the first power supply terminal **24A**, and the supported part is exposed outside the semiconductor device **A10**. On the second terminal base **742** is supported a part of the second power supply terminal **24B**, and the supported part is exposed outside the semiconductor device **A10**. As shown in FIGS. **8** and **13**, a nut **743** is disposed in each of the first terminal base **741** and the second terminal base **742**. Each nut **743** corresponds in the thickness direction z to the coupling hole **241** provided in the first power supply terminal **24A** or the second power supply terminal **24B**. The fastening member such as a bolt inserted in the coupling hole **241** is in threaded engagement with a nut **743**.

(152) As shown in FIGS. **2**, **6** and **9**, the output terminal base **75**, along with the terminal seat **72** connected thereto, supports the output terminal **25**. The output terminal base **75** includes a first terminal base **751** and a second terminal base **752**. The first terminal base **751** and the second terminal base **752** are spaced apart from each other in the first direction x_1 . On the first terminal base **751** is supported a part of the first output terminal **25A**, and the supported part is exposed outside the semiconductor device **A10**. On the second terminal base **752** is supported a part of the second output terminal **25B**, and the supported part is exposed outside the semiconductor device **A10**. As shown in FIGS. **9** and **14**, a nut **753** is disposed in each of the first terminal base **751** and the second terminal base **752**. Each nut **753** corresponds in the thickness direction z to the coupling hole **251** provided in the first output terminal **25A** or the second output terminal **25B**. The fastening member such as a bolt inserted in the coupling hole **251** is in threaded engagement with a nut **753**.

(153) As shown in FIGS. **2**, **11** and **12**, the top plate **79** closes the interior of the semiconductor device **A10** defined by the heat sink **61** and the case **70**. The top plate **79** is supported on the paired side walls **71** of the case **70**, facing the front surface **111** of the substrate **11** and being spaced apart from the front surface **111** in the thickness direction z . The top plate **79** is made of an electrically insulating synthetic resin.

(154) Next, the circuit configuration in the semiconductor device **A10** is described with reference to FIG. **21**.

(155) As shown in FIG. **21**, two switching circuits, namely an upper arm circuit **81** and a lower arm circuit **82** are formed in the semiconductor device **A10**. The upper arm circuit **81** may be made up of the first upper arm mounting layer **211A**, the second upper arm mounting layer **221A**, the third upper arm mounting layer **231A**, and the switching elements **31** and the protective elements **32** electrically bonded to these mounting layers. The switching elements **31** and the protective elements **32** electrically bonded to these mounting layers are connected in parallel between the first

power supply terminal **24A** and the output terminal **25**. The gate electrodes **313** of the switching elements **31** in the upper arm circuit **81** are connected in parallel to the first gate terminal **27A**. The switching elements **31** in the upper arm circuit **81** are driven simultaneously by the application of a gate voltage to the first gate terminal **27A** using a driving circuit such as a gate driver disposed outside the semiconductor device **A10**.

(156) The front surface electrodes **311** of the switching elements **31** in the upper arm circuit **81** are connected in parallel to the first detection terminal **281A**. The source current flowing through the switching elements **31** in the upper arm circuit **81** is inputted to a control circuit disposed outside the semiconductor device **A10** via the first detection terminal **281A**.

(157) In the upper arm circuit **81**, the voltage applied by the first power supply terminal **24A** and the second power supply terminal **24B** to the first upper arm mounting layer **211A**, the second upper arm mounting layer **221A** and the third upper arm mounting layer **231A** is inputted to the control circuit disposed outside the semiconductor device **A10** via the power supply current detection terminal **282**.

(158) The lower arm circuit **82** may be made up of the first lower arm mounting layer **211B**, the second lower arm mounting layer **221B**, the third lower arm mounting layer **231B**, and the switching elements **31** electrically bonded to these mounting layers, and the protective elements **32** are connected in parallel between the output terminal **25** and the second power supply terminal **24B**. The gate electrodes **313** of the switching elements **31** in the lower arm circuit **82** are connected in parallel to the second gate terminal **27B**. The switching elements **31** in the lower arm circuit **82** are driven simultaneously by the application of a gate voltage to the second gate terminal **27B** using a driving circuit such as a gate driver disposed outside the semiconductor device **A10**.

(159) The front surface electrodes **311** of the switching elements **31** in the lower arm circuit **82** are connected in parallel to the second detection terminal **281B**. The source current flowing through the switching elements **31** in the lower arm circuit **82** is inputted to a control circuit disposed outside the semiconductor device **A10** via the second detection terminal **281B**.

(160) Alternating voltages of various frequencies are output from the output terminal **25** by connecting a DC power supply to the first power supply terminal **24A** and the second power supply terminal **24B** and driving the switching elements **31** in the upper arm circuit **81** and the lower arm circuit **82**. The alternating voltage output from the output terminal **25** is supplied to a power supply target such as a motor.

(161) The advantages of the semiconductor device **A10** are described below.

(162) With the configuration of the semiconductor device **A10**, the moisture-resistant layer **51** is held in contact with both of the mounting layer (the first mounting layer **211**, the second mounting layer **221** and the third mounting layer **231**) and the side surface **31C** of the switching elements **31**. In the thickness direction z , the moisture-resistant layer **51** extends to be spanned between the mounting layer and the side surface **31C**. When moisture enters the sealing resin **52** due to the effect of high temperature and high humidity, a leakage current $L_{sub.C}$ is likely to be generated from the front surface electrode **311** of the switching element **31**, as shown in FIG. 22. On the switching element **31**, the leakage current $L_{sub.C}$ tries to flow along the front surface of the insulating film **314** and the side surface **31C**. The provision of the moisture-resistant layer **51** makes the path of the leakage current $L_{sub.C}$ longer, making it difficult for the leakage current $L_{sub.C}$ to flow. Since the leakage current $L_{sub.C}$ is thus prevented from reaching the mounting layer, breakdown of the switching element **31** due to the flow of the leakage current $L_{sub.C}$ is prevented. Thus, the semiconductor device **A10** operates stably under high temperature and high humidity conditions. When the switching element **31** or the protective element **32** has a relatively small thickness of 150 μm or less, the path of the leakage current $L_{sub.C}$ becomes relatively short. When a voltage of 1,200 V or more is applied to such a semiconductor device, the leakage current $L_{sub.C}$ flows relatively easily. Provision of the moisture-resistant layer **51** is particularly effective for such a relatively thin switching element **31**.

(163) In the comparative example B10 shown in FIG. 23, which does not include a moisture-resistant layer 51, the leakage current L.sub.C is conducted to the mounting layer (the first mounting layer 211, the second mounting layer 221 and the third mounting layer 231) along the side surface 31C of the switching element 31. This causes a short circuit between the front surface electrode 311 and the back surface electrode 312 of the switching element 31, resulting in breakdown of the switching element 31.

(164) It is preferable that the moisture-resistant layer 51 contains polyimide. Polyimide is an electrically insulating material that is resistant to temperature cycling and not easily affected by moisture. Thus, by containing polyimide, the moisture-resistant layer 51 reliably prevents the leakage current L.sub.C from flowing along the side surface 31C as shown in FIG. 22 even under high temperature and high humidity conditions.

(165) It is preferable that the moisture-resistant layer 51 contains silicone gel in addition to polyimide. Such a moisture-resistant layer 51 has a lower Young's modulus as compared with a moisture-resistant layer 51 made of polyimide alone. Thus, the moisture-resistant layer 51 easily follows the thermal strain of the switching element 31 during the use of the semiconductor device A10. This reduces the shearing stress acting on the switching element 31.

(166) Moreover, containing polyimide and silicone gel in the moisture-resistant layer 51 enhances the resistance of the moisture-resistant layer 51 to temperature cycling. A semiconductor device A10 with a moisture-resistant layer 51 made of polyimide alone was subjected to a temperature cycling test in a range of -40 to 125° C. As a result, the semiconductor device A10 was broken after about 20 cycles. Presumably, a crack was formed in the moisture-resistant layer 51, and moisture entering through the crack caused the breakdown. A semiconductor device A10 with a moisture-resistant layer 51 made of polyimide and silicone gel was subjected to the same temperature cycling test, and the semiconductor device A10 was not broken even after 1000 cycles. This is because the Young's modulus of this moisture-resistant layer 51 is lower than that of the moisture-resistant layer 51 made of polyimide alone, and hence the shearing stress acting on the moisture-resistant layer 51 due to thermal expansion or contraction is reduced. Thus, it is preferable that the moisture-resistant layer 51 contains polyimide and silicone gel.

(167) The semiconductor device A10 includes wires 41 connected to the front surface electrodes 311 of the switching elements 31, and the wires 41 extend in the first direction x1. Thus, the moisture-resistant layer 51 that covers the side surface 31C of the switching element 31 can be formed without being hindered by the wires 41.

(168) The first bonding portion 411 of each of the paired outer wires 41B has the first connect portion 411A held in contact with the first pad 311A, the second connect portion 411B held in contact with the second pad 311B, and the joint portion 411C located between the first connect portion 411A and the second connect portion 411B. The joint portion 411C projects in the same direction in which the front surface 31A of the switching element 31 faces along the thickness direction z. As shown in FIGS. 16 and 19, it is preferable that the height H of the joint portion 411C in the thickness direction z from the front surface of the front surface electrode 311 of the switching element 31 to the top C of the joint portion 411C is not less than three times the diameter of the wires 41. For example, when the diameter of the wires 41 is $300\text{ }\mu\text{m}$, it is preferable that the height H of the joint portion 411C is $900\text{ }\mu\text{m}$ or more. With such a configuration, the joint portion 411C functions as an elastic member capable of elastically deforming in the first direction x1 to reduce the shearing stress acting on the first connect portion 411A and the second connect portion 411B. Thus, detachment of the first bonding portion 411 from the front surface electrode 311 due to shearing stress is prevented. In the semiconductor device A10, the diameter of the wires 41 is $400\text{ }\mu\text{m}$, and the height H of the joint portion 411C is $1,600\text{ }\mu\text{m}$. Note that when the height H of the joint portion 411C in the semiconductor device A10 is $800\text{ }\mu\text{m}$, at least one of the first connect portion 411A or the second connect portion 411B may be detached in the $\Delta T_{\text{sub.j}}$ power cycling test described later.

(169) The semiconductor device A10 has the heat sink 61 bonded to the back surface 112 of the substrate 11. Thus, the heat generated at the switching elements 31 is efficiently dissipated outside the semiconductor device A10. In this case, it is preferable that the substrate 11 is made of a ceramic with excellent thermal conductivity (e.g. aluminum nitride).

(170) FIGS. 24-50 show semiconductor devices A11 to A15 that are variations of the semiconductor device A10. [First Variation] A semiconductor device A11 according to a first variation of the semiconductor device A10 is described with reference to FIGS. 24 and 25. The semiconductor device A11 is an example in which the contact area of the moisture-resistant layer 51 with the switching element 31 is smaller than that in the foregoing semiconductor device A10. Note that FIG. 24 is a sectional view taken along the same plane as FIG. 16. FIG. 25 is a sectional view taken along the same plane as FIG. 19.

(171) As shown in FIGS. 24 and 25, the moisture-resistant layer 51 covers a part of the side surface 31C of the switching element 31.

(172) The advantages of the semiconductor device A11 are described below.

(173) With the configuration of the semiconductor device A11, the moisture-resistant layer 51 is held in contact with both of the mounting layer (the first mounting layer 211, the second mounting layer 221 and the third mounting layer 231) and the side surface 31C of the switching elements 31. In the thickness direction z, the moisture-resistant layer 51 extends to be spanned between the mounting layer and the side surface 31C. Thus, the semiconductor device A11 also operates stably under high temperature and high humidity conditions.

(174) [Second Variation]

(175) A semiconductor device A12 according to a second variation of the semiconductor device A10 is described with reference to FIGS. 26-31. The semiconductor device A12 is an example in which the contact area of the moisture-resistant layer 51 with the switching element 31 is larger than that in the foregoing semiconductor device A10.

(176) As shown in FIGS. 26-31, at the switching element 31, the moisture-resistant layer 51 is held in contact with both of the side surface 31C and the insulating film 314. The moisture-resistant layer 51 spans the edge 314A, as viewed in the thickness direction z, of the insulating film 314.

(177) The advantages of the semiconductor device A12 are described below.

(178) With the configuration of the semiconductor device A12, the moisture-resistant layer 51 is held in contact with both of the mounting layer (the first mounting layer 211, the second mounting layer 221 and the third mounting layer 231) and the side surface 31C of the switching elements 31. In the thickness direction z, the moisture-resistant layer 51 extends to be spanned between the mounting layer and the side surface 31C. Thus, the semiconductor device A12 also operates stably under high temperature and high humidity conditions.

(179) At the switching element 31 of the semiconductor device A12, the moisture-resistant layer 51 is held in contact with both of the side surface 31C and the insulating film 314. The moisture-resistant layer 51 spans the edge 314A, as viewed in the thickness direction z, of the insulating film 314. Such a configuration makes the path of the leakage current $L_{sub.C}$ shown in FIG. 22 longer than that in the semiconductor device A10, making it more difficult for the leak current $L_{sub.C}$ to flow than in the semiconductor device A10. Moreover, since the moisture-resistant layer 51 covers the insulating film 314, the insulating film 314 is protected from external factors.

(180) [Third Variation]

(181) A semiconductor device A13 according to a third variation of the semiconductor device A10 is described with reference to FIGS. 32-37. The semiconductor device A13 is an example in which the contact area of the moisture-resistant layer 51 with the switching element 31 is larger than that in the foregoing semiconductor device A12.

(182) As shown in FIGS. 32-37, at the switching element 31, the moisture-resistant layer 51 is held in contact with both of the side surface 31C and the insulating film 314. The moisture-resistant layer 51 spans the edge 314A, as viewed in the thickness direction z, of the insulating film 314.

Moreover, the moisture-resistant layer **51** is held in contact with at least a part of the front surface electrode **311**.

(183) The advantages of the semiconductor device **A13** are described below.

(184) With the configuration of the semiconductor device **A13**, the moisture-resistant layer **51** is held in contact with both of the mounting layer (the first mounting layer **211**, the second mounting layer **221** and the third mounting layer **231**) and the side surface **31C** of the switching elements **31**. In the thickness direction z , the moisture-resistant layer **51** extends to be spanned between the mounting layer and the side surface **31C**. Thus, the semiconductor device **A13** also operates stably under high temperature and high humidity conditions.

(185) At the switching element **31** of the semiconductor device **A13**, the moisture-resistant layer **51** is held in contact with both of the side surface **31C** and the insulating film **314**, and also in contact with at least a part of the front surface electrode **311**. As shown in FIGS. **32** and **35**, as viewed in the thickness direction z , the moisture-resistant layer **51** surrounds the front surface electrode **311** while overlapping with a part of the front surface electrode **311**. This configuration improves the dielectric breakdown voltage of the side surface **31C** as compared with that in the semiconductor device **A12**, making it more difficult for the leak current $L_{sub.C}$ to flow than in the semiconductor device **A12**. Moreover, since the moisture-resistant layer **51** covers the insulating film **314**, the insulating film **314** is protected from external factors.

(186) [Fourth Variation]

(187) A semiconductor device **A14** according to a fourth variation of the semiconductor device **A10** is described with reference to FIGS. **38-43**. The semiconductor device **A14** is an example in which the contact area of the moisture-resistant layer **51** with the switching element **31** is larger than that in the foregoing semiconductor device **A13**.

(188) As shown in FIGS. **38-43**, at the switching element **31**, the moisture-resistant layer **51** is held in contact with both of the side surface **31C** and the insulating film **314**. The moisture-resistant layer **51** spans the edge **314A**, as viewed in the thickness direction z , of the insulating film **314**. Moreover, the moisture-resistant layer **51** is held in contact with the front surface electrode **311** and at least a part of the first bonding portions **411** of the wires **41**. Thus, the switching elements **31** are entirely covered with the moisture-resistant layer **51**. However, the first bonding portions **411** are not completely covered with the moisture-resistant layer **51**, and the upper ends of the first bonding portions **411** are exposed from the moisture-resistant layer **51**. That is, the thickness of the moisture-resistant layer **51** covering the front surface **31A** of the switching element **31** is smaller than the diameter of the wires **41**.

(189) As shown in FIGS. **38, 39, 41** and **42**, the moisture-resistant layer **51** covers the entirety of the front surface of the protective element **32** associated with the switching element **31**. However, the second bonding portions **412** of the wires **41** are not completely covered with the moisture-resistant layer **51**, and the upper ends of the second bonding portions **411** are exposed from the moisture-resistant layer **51**. That is, the thickness of the moisture-resistant layer **51** covering the front surface **32A** of the protective element **32** is smaller than the diameter of the wires **41**.

(190) An example of a method for forming the moisture-resistant layer **51** of the semiconductor device **A14** is described below. A liquefied synthetic resin material containing polyimide, silicone gel and solvent is prepared. Note that the solvent is volatile. Then, after the switching elements **31** and the protective elements **32** are electrically connected to the mounting layer on which these elements are mounted (the first mounting layer **211**, the second mounting layer **221** and the third mounting layer **231**), the synthetic resin material is dropped onto respective upper surfaces of a switching element **31** and a protective element **32** with a dispenser. Since the synthetic resin material has fluidity, it spreads over the entirety of the upper surface of the switching element **31** including the front surface electrode **311**, the gate electrode **313** and the insulating film **314**, and further spreads from the side surface **31C** of the switching element **31** onto the mounting layer. Similarly, at the protective element **32**, the entirety of the front surface of the protective element **32**

is covered with the synthetic resin material. Thus, the entirety of the front surface of the switching element **31** is covered with the synthetic resin material. Due to the surface tension of the synthetic resin material, the thickness of the synthetic resin material on the upper surface of the switching element **31** becomes generally uniform. Finally, the synthetic resin material is heat-cured to obtain the moisture-resistant layer **51**. In this process, the solvent volatilizes. With this method, the moisture-resistant layer **51** covering the switching element **31** and the protective element **32** is easily formed.

(191) The advantages of the semiconductor device **A14** are described below.

(192) With the configuration of the semiconductor device **A14**, the moisture-resistant layer **51** is held in contact with both of the mounting layer (the first mounting layer **211**, the second mounting layer **221** and the third mounting layer **231**) and the side surface **31C** of the switching elements **31**. In the thickness direction z , the moisture-resistant layer **51** extends to be spanned between the mounting layer and the side surface **31C**. Thus, the semiconductor device **A14** also operates stably under high temperature and high humidity conditions.

(193) At each of the switching elements **31**, the moisture-resistant layer **51** is held in contact with both of the side surface **31C** and the insulating film **314**. The moisture-resistant layer **51** is also held in contact with the front surface electrode **311** and at least a part of the first bonding portions **411** of the wires **41**. Since the entirety of the switching element **31** is covered with the moisture-resistant layer **51** in this way, moisture entering the sealing resin **52** is prevented from reaching the front surface of the switching element **31**. Thus, dielectric breakdown of the switching elements **31** caused by the leakage current $L_{sub.C}$ shown in FIG. 22, which is due to the influence of moisture, is effectively prevented. Moreover, since the moisture-resistant layer **51** covers the insulating film **314**, the insulating film **314** is protected from external factors.

(194) At each of the switching elements **31**, the paired outer wires **41B**, each of which has the joint portion **411C** of the first bonding portion **411** that projects in the thickness direction z , are arranged on the opposite sides of the paired inner wires **41A** in the second direction x_2 . Thus, the synthetic resin material for forming the moisture-resistant layer **51** can be dropped from above the front surface **31A** of the switching element **31** without interfering with the joint portion **411C**. As shown in FIG. 38, in the semiconductor device **A14**, the joint portions **411C** are provided only in the first bonding portions **411** of the paired outer wires **41B**. It may be considered, as another example, to provide the joint portions **411C** not only in the first bonding portions **411** of the paired outer wires **41B** but also in the first bonding portions **411** of the paired inner wires **41A**. With such a configuration, however, two adjacent joint portions **411C** are located close to each other, which makes it difficult to drop the synthetic resin material for forming the moisture-resistant layer **51** onto the switching element **31**. Moreover, when the synthetic resin material is dropped on the switching element **31**, the synthetic resin material may rise to the upper end of the joint portions **411C**. In such a case, since the Young's modulus of the moisture-resistant layer **51** is relatively high, heat generated from the switching elements **31** exerts a large shearing stress on the joint portions **411C**. This may result in detachment of the first connect portions **411A** and the second connect portions **411B** of the first bonding portions **411** from the front surface electrode **311** of the switching element **31**. Thus, in terms of the reliability of the semiconductor device **A14**, it is preferable that a larger distance is secured between two adjacent joint portions **411C** in the second direction x_2 .

(195) The moisture-resistant layer **51** also covers the entirety of the protective element **32** associated with the switching element **31** which the moisture-resistant layer covers. Thus, protective element **32** is effectively protected from external factors. On the other hand, the first bonding portions **411** are not completely covered with the moisture-resistant layer **51**, and the upper ends of the first bonding portions **411** are exposed from the moisture-resistant layer **51**. That is, the thickness of the moisture-resistant layer **51** covering the front surface **31A** of the switching element **31** is smaller than the diameter of the wires **41**. With this configuration, an excessively

large shearing stress is less likely to act on the first bonding portions **411**, as compared with the configuration in which the first bonding portions **411** are completely covered with the moisture-resistant layer **51**. Thus, detachment of the first bonding portions **411** from the front surface electrode **311** of the switching element **31** is prevented, which enhances the reliability of the semiconductor device **A14**.

(196) Next, with reference to FIG. **50**, description is given of a favorable thickness of the moisture-resistant layer **51** in the semiconductor device **A14**. FIG. **50** shows the results of a H3TRB test and a $\Delta T_{sub.j}$ power cycling test with varying thicknesses of the moisture-resistant layer **51** of the semiconductor device **A14**. The thickness of the moisture-resistant layer **51** shown in FIG. **50** is the thickness at the corner of the insulating film **314** (i.e., the portion connected to both of the edge **314A** and the side surface **31C**) of a switching element **31**. Before the H3TRB test is conducted, the semiconductor device **A14** has been subjected to temperature cycles from -40 to 125°C . The number of the temperature cycles was 300. In the H3TRB test, the semiconductor device **A14** was driven at a DC voltage of 1,360 V as explained later. In the $\Delta T_{sub.j}$ power cycling test, the temperature $\Delta T_{sub.j}$ of the first bonding layer **391** for electrically bonding the switching elements **31** to the mounting layer (the first mounting layer **211**, the second mounting layer **221** and the third mounting layer **231**) was set to 100°C . In light of this, the range of the temperature cycles in the $\Delta T_{sub.j}$ power cycling test was from 50 to 150°C .

(197) The left vertical axis in FIG. **50** is for indicating the withstand time of the semiconductor device **A14** in the H3TRB test. The “withstand time” means the time from the start of the test to the time when dielectric breakdown is observed in at least one of the switching elements **31** of the semiconductor device **A14**. The right vertical axis in FIG. **50** is for indicating the number of temperature cycles done in the $\Delta T_{sub.j}$ power cycling test before the first bonding portion **411** of a wire **41**, connected to the front surface electrode **311** of a switching element **31**, was detached from the front surface electrode **311** (hereinafter referred to as “ $\Delta T_{sub.j}$ power cycle”). The desirable number of temperature cycles (or the standard value of the $\Delta T_{sub.j}$ power cycle, indicated in FIG. **50**) is 15,000 times. The horizontal axis in FIG. **50** represents the thickness of the moisture-resistant layer **51**.

(198) As shown in FIG. **50**, the withstand time of the semiconductor device **A14** increases sharply when the thickness of the moisture-resistant layer **51** exceeds $10\text{ }\mu\text{m}$. This indicates that the resistance to breakdown of the switching element **31** due to moisture intrusion (or reliability related to moisture absorption) improves with increasing thickness of the moisture-resistant layer **51**. On the other hand, the $\Delta T_{sub.j}$ power cycle gradually reduces with increasing thickness of the moisture-resistant layer **51**. This indicates that increasing the thickness of the moisture-resistant layer **51** leads to an increased risk of detachment of the first bonding portion **411** of the wire **41** from the front surface electrode **311** of the switching element **31** or detachment of the second bonding portion **412** of the wire **41** from the anode electrode **321** of the protective element **32**. These test results reveal that preferable thickness of the moisture-resistant layer **51** is in the range from 40 to $200\text{ }\mu\text{m}$. More preferable range of the thickness of the moisture-resistant layer **51** may be from 50 to $100\text{ }\mu\text{m}$. Experiments have confirmed that the thickness of the moisture-resistant layer **51** on the upper surface of the switching element **31** is 1.2 times the thickness of the moisture-resistant layer **51** at the corners. Accordingly, the preferable thickness of the moisture-resistant layer **51** on the upper surface of the switching element **31** is from 48 to $240\text{ }\mu\text{m}$, and more preferably, from 60 to $120\text{ }\mu\text{m}$.

(199) FIG. **51** shows the results (unit: h) of the H3TRB test performed on the semiconductor device **A14** and a comparative example **B10** that does not include the moisture-resistant layer **51** shown in FIG. **23**. As described before, a semiconductor device determined to be acceptable in the H3TRB test (the device withstand time is 1000 h or more) is expected to operate stably under high temperature and high humidity conditions. In the H3TRB test, when the rating voltage is 1,700 V, the DC voltage for driving the semiconductor device **A14** and the comparative example **B10** is set

to 1,360 V (80% of the rating voltage). As a result of the H3TRB test performed based on this DC voltage, the withstand time of the semiconductor device **A14** was found to 1000 h or more, which is acceptable. Thus, the semiconductor device **A14** is expected to operate stably under high temperature and high humidity conditions. On the other hand, the withstand time of the comparative example **B10** was 10 to 500 h, which is not acceptable. The comparative example **B10** is considered to be inferior to the semiconductor device **A14** in terms of the capability of stable operation under high temperature and high humidity conditions.

(200) As shown in FIG. 51, the insulation resistance reduction rate (unit: %) of the sealing resin **52** during the H3TRB test was 20% in the semiconductor device **A14** and 84% in the comparative example **B10**. Presumably, the insulation resistance reduction rate of the sealing resin **52** shown in FIG. 51 were obtained because even when moisture enters the sealing resin **52** due to a high temperature and high humidity environment, the moisture-resistant layer **51** hinders the leakage current $L_{sub.C}$ shown in FIG. 22 from flowing along the side surfaces **31C** of the switching element.

(201) [Fifth Variation]

(202) A semiconductor device **A15** according to a fifth variation of the semiconductor device **A10** is described with reference to FIGS. 44-49. The semiconductor device **A15** is an example in which the thickness of the moisture-resistant layer **51** on the upper surface of a switching element **31** is larger than that in the foregoing semiconductor device **A14**.

(203) As shown in FIGS. 44-49, at the switching elements **31**, the moisture-resistant layer **51** covers both of the switching elements **31** and the first bonding portions **411** of the wires **41**.

(204) As shown in FIGS. 44, 45, 47 and 48, the moisture-resistant layer **51** covers the entirety of the front surface of the protective element **32** associated with the switching element **31** and the second bonding portions **412** of the wires **41** connected to the anode electrode **321** of the protective element **32**.

(205) The advantages of the semiconductor device **A15** are described below.

(206) With the configuration of the semiconductor device **A15**, the moisture-resistant layer **51** is held in contact with both of the mounting layer (the first mounting layer **211**, the second mounting layer **221** and the third mounting layer **231**) and the side surface **31C** of switching elements **31**. In the thickness direction z , the moisture-resistant layer **51** extends to be spanned between the mounting layer and the side surface **31C**. Thus, the semiconductor device **A15** also operates stably under high temperature and high humidity conditions.

(207) At the switching element **31**, the moisture-resistant layer covers both of the switching element **31** and the first bonding portions **411** of the wires **41**. Since the entirety of the switching element **31** is covered with the moisture-resistant layer **51** in this way, moisture entering the sealing resin **52** is prevented from reaching the front surface **31A** of the switching element **31**. Thus, dielectric breakdown of the switching elements **31** caused by the leakage current $L_{sub.C}$ shown in FIG. 22, which is due to the influence of moisture, is effectively prevented. Moreover, since the moisture-resistant layer **51** covers the insulating film **314**, the insulating film **314** is protected from external factors. In the semiconductor device **A15** again, it is preferable that the thickness of the moisture-resistant layer **51** on the upper surface of the switching element **31** is from 48 to 240 μm .

Second Embodiment

(208) A semiconductor device **A20** according to a second embodiment of the present disclosure is described below with reference to FIGS. 52-57. In these figures, the elements that are identical or similar to those of the foregoing semiconductor device **A10** are designated by the same reference signs as those used for the foregoing embodiment and, descriptions thereof are omitted.

(209) The semiconductor device **A20** differs from the foregoing semiconductor device **A10** in that it includes clips **47** instead of the wires **41**.

(210) As shown in FIGS. 52-54, the clips **47** are electrically bonded to the front surface electrodes **311** of the switching elements **31** and the first lower arm mounting layer **211B**, the second lower

arm mounting layer **221B** or the third lower arm mounting layer **231B**. As shown in FIGS. 55-57, the clips **47** are electrically bonded to the front surface electrodes **311** of the switching elements **31** and the first electroconductive layer **212**, the second electroconductive layer **222** or the third electroconductive layer. The clips **47** are formed by bending a thin metal plate such as a copper plate. As shown in FIGS. 52 and 55, the clips **47** are each in the form of a strip extending in the first direction x_1 as viewed in the thickness direction z . As shown in FIGS. 53 and 56, the clips **47** have a hook-like shape as viewed in the second direction x_2 . As shown in FIGS. 53 and 56, the clips **47** are electrically bonded to an object such as the front surface electrode **311** by using a clip bonding layer **49**. The clip bonding layer **49** is electrically conductive. For example, the clip bonding layer **49** is made of lead-free solder mainly composed of tin. To use the clip bonding layer **49**, a plating layer of nickel or gold, for example, is applied to the front surface of the front surface electrode **311**. In the semiconductor device **A20**, when the front surface electrode **311** is covered with the moisture-resistant layer **51**, the clip bonding layer **49** and the plating layer are also covered with the moisture-resistant layer **51**.

(211) As shown in FIGS. 52-54, in the switching elements **31** electrically bonded to the first upper arm mounting layer **211A**, the clips **47** are electrically bonded to the front surface electrodes **311** and the first lower arm mounting layer **211B**. As shown in FIGS. 55-57, in the switching elements **31** electrically bonded to the first lower arm mounting layer **211B**, the clips **47** are electrically bonded to the front surface electrodes **311** and the first electroconductive layer **212**. Thus, the front surface electrodes **311** of the switching elements **31** electrically bonded to the first mounting layer **211** are electrically connected to the first lower arm mounting layer **211B** or the first electroconductive layer **212**.

(212) As shown in FIGS. 52-54, in the switching elements **31** electrically bonded to the second upper arm mounting layer **221A**, the clips **47** are electrically bonded to the front surface electrodes **311** and the second lower arm mounting layer **221B**. As shown in FIGS. 55-57, in the switching elements **31** electrically bonded to the second lower arm mounting layer **221B**, the clips **47** are electrically bonded to the front surface electrodes **311** and the second electroconductive layer **222**. Thus, the front surface electrodes **311** of the switching elements **31** electrically bonded to the second mounting layer **221** are electrically connected to the second lower arm mounting layer **221B** or the second electroconductive layer **222**.

(213) As shown in FIGS. 52-54, in the switching elements **31** electrically bonded to the third upper arm mounting layer **231A**, the clips **47** are electrically bonded to the front surface electrodes **311** and the third lower arm mounting layer **231B**. As shown in FIGS. 55-57, in the switching elements **31** electrically bonded to the third lower arm mounting layer **231B**, the clips **47** are electrically bonded to the front surface electrodes **311** and the third electroconductive layer **232**. Thus, the front surface electrodes **311** of the switching elements **31** electrically bonded to the third mounting layer **231** are electrically connected to the third lower arm mounting layer **231B** or the third electroconductive layer **232**.

(214) With reference to FIGS. 52 and 53, description is given below of the configuration of the clips **47** for each of the switching elements **31** electrically bonded to the first upper arm mounting layer **211A**, the second upper arm mounting layer **221A** or the third upper arm mounting layer **231A**. As shown in FIG. 53, each clip **47** is electrically bonded also to the anode electrode **321** of the protective element **32** associated with the switching element **31** by using the clip bonding layer **49**. Thus, the anode electrode **321** of the protective element **32** electrically bonded to the first upper arm mounting layer **211A** is electrically connected to both of the front surface electrode **311** of the corresponding switching element **31** and the first lower arm mounting layer **211B**. The anode electrode **321** of the protective element **32** electrically bonded to the second upper arm mounting layer **221A** is electrically connected to both of the front surface electrode **311** of the corresponding switching element **31** and the second lower arm mounting layer **221B**. Also, the anode electrode **321** of the protective element **32** electrically bonded to the third upper arm mounting layer **231A** is

electrically connected to both of the front surface electrode **311** of the corresponding switching element **31** and the third lower arm mounting layer **231B**.

(215) As shown in FIGS. **52** and **53**, each of the clips **47** has an opening **471** penetrating in the thickness direction z . The opening **471** is located between the front surface electrode **311** of the switching element **31** and the anode electrode **321** of the protective element **32** in the first direction x . As viewed in the thickness direction z , the edge **314A** of the insulating film **314** of the switching element **31** is visible through the opening **471**. When clips **47** are electrically bonded to the front surface electrodes **311** of the switching elements **31**, most of the switching elements **31** are covered with the clips **47**. By forming an opening **471** in each clip **47** at a location overlapping with the switching element **31** as viewed in the thickness direction, the synthetic resin material for forming the moisture-resistant layer **51** can be dropped under the clip **47**. Thus, the synthetic resin material can be dropped uniformly over the entirety of the switching element **31**. Although the semiconductor device **A20** is described as having an opening **471** in each clip **47**, a cutout penetrating in the thickness direction z may be formed, instead of the opening **471**, in each clip **46** at a location overlapping with the switching element **31** as viewed in the thickness direction z .

(216) With reference to FIGS. **55** and **56**, description is given below of the configuration of the clips **47** for each of the switching elements **31** electrically bonded to the first lower arm mounting layer **211B**, the second lower arm mounting layer **221B** or the third lower arm mounting layer **231B**. As shown in FIG. **56**, each clip **47** is electrically bonded also to the anode electrode **321** of the protective element **32** associated with the switching element **31** by using the clip bonding layer **49**. Thus, the anode electrode **321** of the protective element **32** electrically bonded to the first lower arm mounting layer **211B** is electrically connected to both of the front surface electrode **311** of the corresponding switching element **31** and the first electroconductive layer **212**. The anode electrode **321** of the protective element **32** electrically bonded to the second lower arm mounting layer **221B** is electrically connected to both of the front surface electrode **311** of the corresponding switching element **31** and the second electroconductive layer **222**. Also, the anode electrode **321** of the protective element **32** electrically bonded to the third lower arm mounting layer **231B** is electrically connected to both of the front surface electrode **311** of the corresponding switching element **31** and the third electroconductive layer **232**. Since the clip **47** is electrically bonded to the anode electrode **321**, the paired auxiliary wires **46** shown in FIG. **18** are not connected to the anode electrode **321** in the semiconductor device **A20**.

(217) As shown in FIGS. **55** and **56**, each of the clips **47** has a pair of openings **471** penetrating in the thickness direction z . The paired openings **471** are located on the opposite sides of the front surface electrode **311** of the switching element **31** in the first direction x_1 . As viewed in the thickness direction z , the edge **314A** of the insulating film **314** of the switching element **31** is visible through the paired openings **471**.

(218) The advantages of the semiconductor device **A20** are described below.

(219) With the configuration of the semiconductor device **A20**, the moisture-resistant layer **51** is held in contact with both of the side surfaces **31C** of the switching elements **31** and the first mounting layer **211**, the second mounting layer **221** or the third mounting layer **231**. In the thickness direction, the moisture-resistant layer **51** extends to be spanned between the first mounting layer **211**, the second mounting layer **221** or the third mounting layer **231** and the side surfaces **31C**. Thus, the semiconductor device **A20** also operates stably under high temperature and high humidity conditions.

(220) The configuration of the moisture-resistant layer **51** of the semiconductor device **A20** is the same as that of the semiconductor device **A10**. Note however that the configuration of the moisture-resistant layer **51** in the semiconductor devices **A11** to **A15** may be employed in the semiconductor device **A20**.

(221) The semiconductor device **A20** has clips **47** instead of the wires **41**. The cross sectional area (the area in cross section along the second direction x_2) of the clip **47** is larger than that of the

wires **41**. Thus, the electric resistance of the clip **47** is lower than that of the wires **41**. Thus, the parasitic resistance of the semiconductor device **A20** is lower than that of the semiconductor device **A10**, so that the power loss of the semiconductor device **A20** is reduced as compared with the semiconductor device **A10**.

(222) Since the cross sectional area of the clip **47** is larger than that of the wires **41**, the clip **47** conducts more heat in the first direction **x1** than the wires **41**. Thus, heat generated from the switching elements **31** is dissipated more efficiently. For example, on the first substrate **11A**, the heat generated from the switching elements **31** is likely to be accumulated in the first upper arm mounting layer **211A** that constitutes the upper arm circuit **81** shown in FIG. **21**. The clips **47** efficiently dissipate the heat accumulated in the first upper arm mounting layer **211A** to the first lower arm mounting layer **211B** and the first electroconductive layer **212**.

(223) The number of the switching elements electrically bonded to each of the first mounting layer **211**, the second mounting layer **221** and the third mounting layer **231** can be set appropriately in accordance with the required power conversion. The first mounting layer **211**, the second mounting layer **221** and the third mounting layer **231** are examples of a “mounting layer” as set forth in the appended claims of the present disclosure. The number of sections constituting the “mounting layer” is not limited to six as is in the present disclosure, and may be set appropriately.

(224) The foregoing embodiments show the example in which the moisture-resistant layer **51** covers the switching elements **31** and the protective elements **32** connected anti-parallel to the switching elements **31**. However, in a semiconductor device that does not use the protective elements **32** but uses the switching elements **31** alone (the configuration that does not use an external free wheeling diode), the moisture-resistant layer **51** may cover the switching elements **31**. Further, the present disclosure is applicable not only to switching elements but also to rectifier elements. For example, the present disclosure is applicable to a semiconductor device that includes a plurality of schottky-barrier diodes. In this case, the configuration such as the material, thickness or formation area of the moisture-resistant layer **51** is the same as that of the foregoing embodiments.

(225) In the foregoing embodiments, as one example, the semiconductor device includes a substrate **11** on which electrically conductive members (the mounting layer and the electroconductive layer) made of a thin metal film are disposed, and the switching elements **31** electrically bonded to the electrically conductive members. The present disclosure is not limited to such an example and also applicable to a resin package-type semiconductor device that includes a lead frame on which elements such as switching elements or rectifier elements are electrically bonded and resin-molded. Since such a semiconductor device also has a risk of moisture intrusion through the sealing resin, covering the entire surfaces or side surfaces of the switching elements or rectifier elements with the moisture-resistant layer according to the present disclosure provides the same advantages. Note that the connection structure by wire bonding using the wires **41** in the semiconductor device **A10** or the connection structure using a thin metal plate or clips **47** in the semiconductor device **A20** is also applicable to a resin package-type semiconductor device.

(226) The present disclosure is not limited to foregoing embodiments. The specific configuration of each part of the present disclosure may be varied in many ways.

(227) The present disclosure at least includes the configurations related to the following clauses in addition to the configurations as set forth in the appended claims.

(228) Clause 1.

(229) A semiconductor device comprising: a first electroconductive layer; a second electroconductive layer spaced apart from the first electroconductive layer; a semiconductor element including a semiconductor layer, a front surface electrode provided on an upper surface of the semiconductor layer, and a back surface electrode provided on a lower surface of the semiconductor layer, the semiconductor element being mounted on the first electroconductive layer with the back surface electrode electrically connected to the first electroconductive layer; a

connection structure electrically connected to the front surface electrode and the second electroconductive layer; a first insulating layer covering at least a side surface of the semiconductor element; and a second insulating layer covering the first insulating layer, wherein the first insulating layer is made of a material having a lower moisture permeability than the second insulating layer. The first insulating layer functions as a barrier film for preventing moisture intrusion.

Clause 2.

(230) The semiconductor device according to clause 1, wherein the first insulating layer covers an entirety of the semiconductor element.

(231) Clause 3.

(232) The semiconductor device according to clause 1, wherein the connection structure includes a connecting portion held in contact with the front surface electrode, and the first insulating layer covers an entirety of the semiconductor element except the connecting portion.

Clause 4.

(233) The semiconductor device according to clause 1, wherein the first insulating layer is from 40 to 200 μm in thickness at a corner located between an upper surface and the side surface of the semiconductor element, and the first insulating layer is from 48 to 240 μm in thickness on the upper surface of the semiconductor element.

Clause 5.

(234) The semiconductor device according to clause 4, wherein the first insulating layer is from 50 to 100 μm in thickness at the corner located between the upper surface and the side surface of the semiconductor element, and the first insulating layer is from 60 to 120 μm in thickness on the upper surface of the semiconductor element.

Clause 6.

(235) The semiconductor device according to clause 1, wherein the connection structure includes a connection structure using a wire, and a thickness of the first insulating layer on an upper surface of the semiconductor element is smaller than a diameter of the wire.

Clause 7.

(236) The semiconductor device according to clause 6, wherein the wire includes a connecting portion held in contact with the front surface electrode, and the thickness of the first insulating layer on the upper surface of the semiconductor element is smaller than a height of the connecting portion (a distance from a front surface of the front surface electrode to a top of the connecting portion). That is, the top of the connecting portion is exposed from the first insulating layer. The connecting portion is crushed with a wedge tool during a bonding process, and the height of the connecting portion is smaller than the diameter of the wire.

Clause 8.

(237) The semiconductor device according to clause 1, wherein the semiconductor layer is 400 μm or less in thickness.

(238) Clause 9.

(239) The semiconductor device according to clause 8, wherein the semiconductor layer is 150 μm or less in thickness.

(240) Clause 10.

(241) The semiconductor device according to clause 1, wherein the semiconductor element further comprises a voltage withstanding structure including an insulating layer covering an upper surface of the semiconductor layer and surrounding an edge of the front surface electrode, and the first insulating layer covers the voltage withstanding structure. In the voltage withstanding structure, an oxide film or a nitride film is formed on the semiconductor layer, and a layer such as a polyimide layer or a polybenzoxazole layer is formed thereon as the insulating layer.

(242) Clause 11.

(243) The semiconductor device according to clause 1, wherein the first insulating layer contains a

synthetic resin that is polyimide or polybenzoxazole.

(244) Clause 12.

(245) The semiconductor device according to clause 11, wherein the first insulating layer contains silicone gel.

(246) Clause 13.

(247) The semiconductor device according to clause 12, wherein the synthetic resin and the silicone gel are uniformly dispersed in the first insulating layer.

(248) Clause 14.

(249) The semiconductor device according to clause 12 or 13, wherein a content by weight of the synthetic resin in the first insulating layer is larger than that of the silicone gel.

(250) Clause 15.

(251) The semiconductor device according to clause 14, wherein the ratio of content by weight of the synthetic resin to the silicone gel in the first insulating layer is from 1.5:1 to 7.0:1.

(252) Clause 16.

(253) The semiconductor device according to clause 1, wherein the connection structure includes a connection structure using a wire and a connection structure using a thin metal plate.

(254) Clause 17.

(255) The semiconductor device according to clause 1, wherein the semiconductor layer is made of a semiconductor material mainly composed of silicon carbide.

(256) Clause 18

(257) The semiconductor device according to clause 17, wherein the semiconductor element includes a MOSFET or a schottky-barrier diode.

(258) Clause 19.

(259) The semiconductor device according to clause 1, wherein breakdown voltage of the protective element is 1,200 V or more.

(260) Clause 20.

(261) The semiconductor device according to clause 1, wherein the first electroconductive layer and the second electroconductive layer are made of a lead frame, and the second insulating layer comprises a resin package sealing the first electroconductive layer, the second electroconductive layer, the semiconductor element and the connection structure.

Clause 21.

(262) The semiconductor device according to clause 1, wherein the first electroconductive layer and the second electroconductive layer comprise a metal layer disposed on an insulating substrate, and the second insulating layer includes a resin package sealing the insulating substrate, the first electroconductive layer, the second electroconductive layer, the semiconductor element and the connection structure. The sealing resin contains silicone gel.

Clause 22.

(263) The semiconductor device according to clause 1, wherein the second insulating layer has a front surface exposed to outside air (atmosphere).

(264) Clause 23.

(265) A method for manufacturing a semiconductor device as set forth in any one of clauses 1 to 22, the method comprising the steps of: preparing a synthetic resin material containing: a material having a lower moisture permeability than a material forming the second insulating layer, or a precursor thereof; and a volatile solvent, electrically connecting the back surface electrode to the first electroconductive layer, dropping the synthetic resin material onto an upper surface of the semiconductor element to cover the semiconductor element with the synthetic resin material, and heat-curing the synthetic resin material with the semiconductor element covered with the synthetic resin material, to thereby form the first insulating layer. Before the step of heat-curing the synthetic resin material, the synthetic resin material does not need to have the function of the first insulating layer. It is only necessary that the synthetic resin material has the function of the first insulating

layer after it is heat-cured. For example, polyimide is dissolved in the solvent in the state of a precursor, and it becomes polyimide through “imidization” after heat-curing, to thereby have the function as the first insulating layer.

Clause 24.

(266) The method for manufacturing the semiconductor device according to clause 23, further comprising the step of connecting the connection structure to the front surface electrode and the second electroconductive layer before the step of covering the semiconductor element with the synthetic resin material.

Claims

1. A semiconductor device comprising: a first mounting layer including an upper surface facing a vertical direction upwardly; a bonding layer that is electrically conductive and disposed on the upper surface of the first mounting layer; a switching element which includes a first element upper surface facing the vertical direction upwardly, a first element lower surface facing an opposite direction of the first element upper surface, and a first element side surface connected to both of the first element upper surface and the first element lower surface, the switching element being electrically bonded to the first mounting layer with the first element lower surface facing the upper surface of the first mounting layer; a conductive member connected to an upper surface electrode of the first element upper surface; a resin layer that covers at least the first element side surface; and a sealing resin that covers the switching element, the resin layer, and at least a part of the conductive member, wherein the resin layer is held in contact with both of a side surface of the bonding layer and the first element side surface in a manner such that the resin layer straddles the side surface of the bonding layer and the first element side surface in a cross section along the vertical direction.
2. The semiconductor device according to claim 1, wherein the side surface of the bonding layer is flush with a side surface of the switching element at their joint.
3. The semiconductor device according to claim 1, further comprising a second mounting layer that is electrically connected to the conductive member, wherein the resin layer does not cover the second mounting layer.
4. The semiconductor device according to claim 1, wherein the resin layer does not cover the first element upper surface.
5. The semiconductor device according to claim 1, wherein the resin layer covers at least a part of the first element upper surface.
6. The semiconductor device according to claim 1, wherein the switching element is a MOSFET formed on a SiC substrate.
7. The semiconductor device according to claim 1, wherein the first mounting layer comprises a lead frame.
8. The semiconductor device according to claim 1, further comprising a substrate on a surface of a conductive layer opposite to a surface of the conductive layer on which the switching element is mounted.
9. The semiconductor device according to claim 8, wherein the substrate is made of a ceramic with excellent thermal conductivity.
10. The semiconductor device according to claim 8, wherein the resin layer is disposed on the first mounting layer, and is not in contact with the substrate.
11. The semiconductor device according to claim 8, wherein the sealing resin does not cover a side of the substrate.
12. The semiconductor device according to claim 8, wherein a side surface and a lower surface of the substrate are exposed from the sealing resin.
13. The semiconductor device according to claim 1, wherein the conductive member comprises

wires.

14. The semiconductor device according to claim 1, wherein the conductive member comprises a metal plate.

15. The semiconductor device according to claim 1, further comprising protective elements parallelly connected to the switching element, wherein the resin layer is arranged between the protective elements.

16. The semiconductor device according to claim 1, wherein the resin layer comprises polyimide.

17. The semiconductor device according to claim 15, wherein the resin layer comprises silicone gel.

18. The semiconductor device according to claim 15, wherein the resin layer covers the protective elements.

19. A semiconductor device comprising: a first lead frame including an upper surface facing a vertical direction upwardly; a bonding layer that is electrically conductive and disposed on the upper surface of the first lead frame; a switching element which includes a first element upper surface facing the vertical direction upwardly, a first element lower surface facing an opposite direction of the first element upper surface, and a first element side surface connected to both of the first element upper surface and the first element lower surface, the switching element being electrically bonded to the first lead frame with the first element lower surface facing the upper surface of the first lead frame; a conductive member connected to an upper surface electrode of the first element upper surface; a resin layer that covers at least the first element side surface; and a sealing resin that covers the switching element, the resin layer, and at least a part of the conductive member, wherein the resin layer is held in contact with both of a side surface of the bonding layer and the first element side surface in a manner such that the resin layer straddles the side surface of the bonding layer and the first element side surface in a cross section along the vertical direction.

20. The semiconductor device according to claim 19, wherein the side surface of the bonding layer is flush with a side surface of the switching element at their joint.

21. The semiconductor device according to claim 19, further comprising a second lead frame that is electrically connected to the conductive member, wherein the resin layer does not cover the second lead frame.

22. The semiconductor device according to claim 19, wherein the resin layer does not cover the first element upper surface.

23. The semiconductor device according to claim 19, wherein the resin layer covers at least a part of the first element upper surface.

24. The semiconductor device according to claim 19, wherein the switching element is a MOSFET formed on a SiC substrate.

25. The semiconductor device according to claim 19, wherein the conductive member comprises wires.

26. The semiconductor device according to claim 19, wherein the conductive member comprises a thin metal plate.

27. The semiconductor device according to claim 19, further comprising protective elements parallelly connected to the switching element, wherein the resin layer is arranged between the protective elements.

28. The semiconductor device according to claim 19, wherein the resin layer comprises polyimide.

29. The semiconductor device according to claim 19, wherein the resin layer comprises silicone gel.

30. The semiconductor device according to claim 27, wherein the resin layer covers the protective elements.
